

7

PPE and Intangibles

**SPOTLIGHT**

| Airbus Group

www.airbusgroup.com

The International Air Traffic Association (IATA) reported 3.7 billion commercial airline passenger departures in 2016. That averaged out to be more than 10 million passengers flying each day! Many of them traveled on Airbus planes, from the A320 family of aircrafts to the world's largest commercial aircraft, the A380. Airbus delivers almost 700 aircrafts a year to its customers. What does it take to manufacture those aircrafts? ●

Alexey Y. Petrov/Shutterstock



To be able to operate on such a scale, Airbus has to decide where to allocate its assets, from cash and receivables (Chapter 5), inventory (Chapter 6), to long-term assets such as Property, Plant and Equipment (PPE), as well as intangible assets. Airbus Group's consolidated Balance Sheet and extracts of its PPE and intangible assets are shown below. Property, Plant, and Equipment and intangible assets form about 30% of Airbus Group's total assets.

A1				
	A	B	C	D
1	Airbus Group			
2	Consolidated Balance Sheet (Adapted)			
3	At December 31			
4	(In millions of Euro)	2015	2014	
5	Current assets	53,243	47,682	
6	Property, plant and equipment	17,127	16,321	
7	Intangible assets	12,555	12,758	
8	All other non-current assets	23,756	19,341	
9	Total assets	106,681	96,102	
10	Total liabilities	100,708	89,023	
11	Total equity	5,973	7,079	
12	Total liabilities and equity	106,681	96,102	
13				
14	Notes to the accounts:			
15	Property, plant and equipment (line 4) 31 Dec 2015	Cost	ADI*	Carrying Amount
16	Land, leasehold improvements and buildings	9,518	(4,349)	5,169
17	Technical equipment and machinery	20,296	(11,946)	8,350
18	Other equipment, factory and office equipment	4,324	(3,290)	1,034
19	Construction in progress	2,574	–	2,574
20	Total property, plant, and equipment	36,712	(19,585)	17,127
21				
22	Intangible assets (line 5) 31 Dec 2015	Cost	AAI*	Carrying Amount
23	Goodwill	10,995	(1,088)	9,907
24	Capitalized development costs	2,686	(1,027)	1,659
25	Other intangible assets	3,375	(2,386)	989
26	Total intangible assets	17,056	(4,501)	12,555
27				

ADI* = Accumulated Depreciation and Impairment; AAI* = Accumulated Amortization and Impairment

PPE and intangibles are long-term assets because they provide economic benefits that extend beyond a single financial period. The allocation of their costs over their useful lives is called depreciation (for PPE) or amortization (for intangible assets). This chapter will start with an overview of various types of long-term assets that businesses have in their operations before proceeding to discuss the specific accounting treatments for long-term assets. As different companies have different types of long-term assets, we will use a few companies as illustrations during our discussions.

LEARNING OBJECTIVES

- | | |
|--|--|
| <ol style="list-style-type: none"> 1 Understand the different types of long-term assets 2 Determine the cost of PPE on initial recognition 3 Understand when to capitalize or expense subsequent costs 4 Measure and record depreciation | <ol style="list-style-type: none"> 5 Account for PPE disposals 6 Understand the recognition and subsequent measurement of intangible assets 7 Evaluate a company's performance based on its assets |
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UNDERSTAND THE DIFFERENT TYPES OF LONG-TERM ASSETS

- 1 Understand** the different types of long-term assets

Property, Plant and Equipment (PPE)

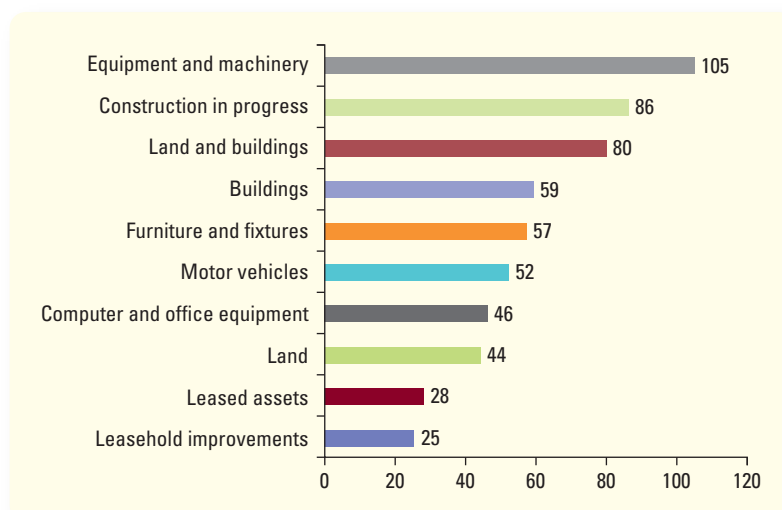
Property, Plant and Equipment (**PPE**), sometimes called *fixed assets*, are long-term, non-current or long-lived assets that are tangible—for instance, land, buildings, and equipment. They may be held for use in the production or supply of goods or services, for rental to others, or for administrative purposes, and they are expected to be used during more than one period. The allocation of a PPE's cost over its useful life is called *depreciation*. The amount that has been allocated over the years is called *accumulated depreciation*. The primary source of guidance for accounting for PPE is *IAS 16—Property, Plant and Equipment*.

Businesses use several types of PPE, as shown in Airbus's Balance Sheet. It has PPE totaling €17,127 million (line 4), which is further detailed in its notes to the accounts (lines 13–17) into (1) land, leasehold improvements and buildings; (2) technical equipment and machinery; (3) other equipment, factory and office equipment; and (4) construction in progress, each with its own costs and accumulated depreciation and **impairment**. The difference between a PPE's cost and its accumulated depreciation is called the “carrying amount” (or net book value). Note that the carrying amount on the Balance Sheet tallies with the details provided in the notes to the accounts (line 4 and line 18).

Different entities may classify their PPE items into somewhat different categories that are suitable (and meaningful) for their business and financial statement users. An airline company, such as Singapore Airlines, would typically use additional PPE categories such as “Aircraft” and “Aircraft spare parts.” Hutchison Whampoa, a Hong Kong-based diversified company with interests in the telecommunication industry amongst many others, uses a PPE category called “telecommunication network assets.” Sinopec (China's largest oil producer and refiner) uses a PPE category “Oil depots, storage tanks, and service stations.”

A recent survey of 170 IFRS companies showed the following top 10 PPE headings reported by these companies, as shown in Exhibit 7-1.

Exhibit 7-1 | Top 10 PPE Categories



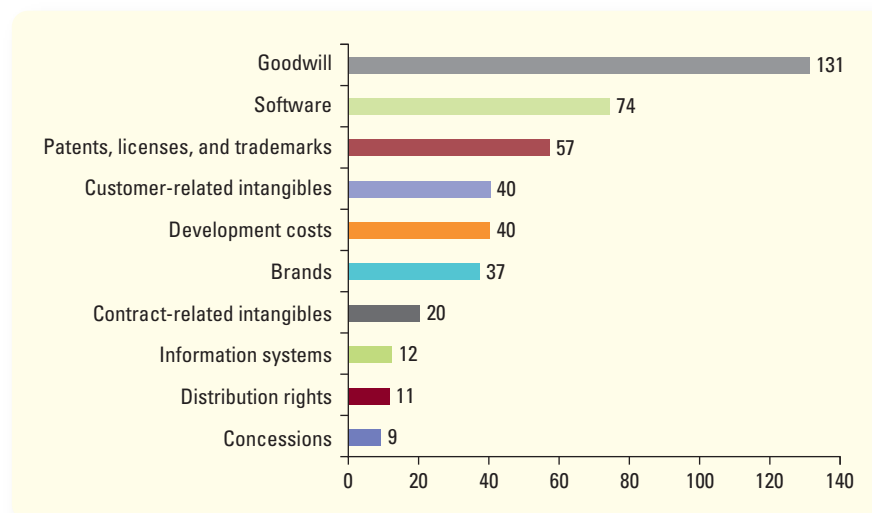
Intangible Assets

Intangible assets are identifiable non-monetary assets without physical substance. Non-monetary simply means that the asset is not expressed in fixed or determinable amounts of money. These intangible assets are unique because they do not have any physical form. Airbus reports a total of €12,555 million of net intangible assets (line 5), comprising (1) goodwill, (2) capitalized development costs, and (3) other intangible assets (lines 21–23). You can check that line 5 tallies with line 24, too. Accounting for intangibles is similar in nature to accounting for PPE assets. With the exception of goodwill (and other intangible assets with indefinite useful lives, more on this later), the costs of intangible assets are also allocated over the assets' respective useful lives. We usually refer to this as amortization. The primary source of guidance for intangible assets is *IAS 38—Intangible Assets*.

Similarly, the categories of intangible assets differ between entities depending on what an entity actually has. For example, Lenovo, the world's second largest computer vendor, categorizes its \$2.1 billion intangible assets into (1) goodwill, (2) trademarks and trade names, (3) internal use software, (4) customer relationships, and (5) patent and technology.

A recent survey of 170 IFRS companies showed that the majority of them (77%) have goodwill on their Balance Sheets (AICPA, 2010). Other intangible assets reported include software, patents, licenses, trademarks, development costs, customer lists and relationships, and many others. Exhibit 7-2 shows the top 10 intangible assets reported by these companies.

Exhibit 7-2 | Top 10 Intangible Asset Categories



Both PPE and intangibles are subject to impairment tests (*IAS 36—Impairment of Assets*) to ensure that the values reported on the Balance Sheet do not exceed the fair value of the assets. Accounting for PPE and intangibles has its own terminology. Different names apply to the individual PPE categories and their corresponding expenses, as shown in Exhibit 7-3.

Other Non-Current Assets

- Some entities, usually property developer and contract manufacturers, have a *Construction in Progress* non-current asset. You saw earlier in Exhibit 7-1 that construction in progress is quite common amongst the companies surveyed in the AICPA (2010) study. This account is a placeholder to hold the costs incurred for assets under construction. Once completed, the cost of the asset that has been accumulated in the Construction in Progress account is then moved to the PPE (or Intangible Asset, if appropriate) account if it is to be used internally. In cases where assets being constructed are meant for sale or delivery to customers, they will be transferred to inventory (and then subsequently to cost of sales). Accounting for long-term construction contracts is beyond the scope of this course, but you can refer to *IAS 11—Construction Contracts* if you want to find out more.

Exhibit 7-3 | PPE and Intangibles Terminology

Asset Account (Balance Sheet)	Related Expense Account (Income Statement)
Property, Plant, and Equipment	
Freehold Land	None
Leasehold Land	Depreciation
Buildings, Machinery and Equipment	Depreciation
Furniture and Fixtures	Depreciation
Land Improvements	Depreciation
Natural Resources	Depletion
Intangibles	
Intangibles with finite useful lives	Amortization
Intangibles with indefinite useful lives	None

- Sometimes, you may also see entities, such as real estate companies or hotels, with *Investment Properties* as a non-current asset. These are a specially designated class of properties (land and/or buildings) held to earn rentals or for capital appreciation or both, rather than for usage associated with sales, production, or general administrative functions. Investment properties are beyond the scope of this course, but you can refer to *IAS 40—Investment Properties* for additional information.
- You may also see companies reporting “Lease Asset” on their Balance Sheets, as well as a corresponding “Lease Liabilities.” These are assets and liabilities that are recognized on the financial statement in relation to lease arrangements. We will discuss these later in Chapter 9.
- For certain companies in the agriculture industry, you may also see a category labeled “biological assets.” For example, Qian Hu (an ornamental fish-breeder headquartered in Singapore) has biological assets that are for sale (as inventory) as well as biological assets that are for breeding purposes (and thus depreciated!). Illovo, Africa’s biggest sugar producer, has cane roots and growing cane as their agriculture assets. JBS Group (the world’s largest meat producer, headquartered in São Paulo, Brazil) has “cattle, hogs and lamb, poultry and plants for harvests” as its biological assets. Agriculture and biological assets are accounted for under *IAS 41—Agriculture*.

Now that you have been introduced to common types of PPE and intangibles, let’s see how we can recognize and measure them.

DETERMINE THE COST OF PPE ON INITIAL RECOGNITION

Recognition of PPE and Intangible Assets

Property, Plant, and Equipment and intangible assets are recognized in financial statements using the same way as other assets, when (1) it is probable that future economic benefits associated with the item will flow to the entity and (2) the cost of the item can be measured reliably. We will start our discussion with PPE for now and discuss intangible assets later.

2 **Determine** the cost of PPE on initial recognition

Measurement of PPE on Initial Recognition

Here is a basic working rule for determining the cost of an asset: *The cost of any asset is the sum of all the costs incurred to bring the asset to its intended use.* Specifically, *IAS 16* requires that the cost of an item of PPE includes the following:

- Its purchase price, including import duties and non-refundable purchase taxes, after deducting trade discounts and rebates.
- Any costs directly attributable to bringing the asset to the location and condition necessary for it to be capable of operating in the manner intended by management.



A Closer Look

There is a third cost element in *IAS 16* that is seldom applicable to the majority of PPE items. *IAS 16* requires that the cost of an item of PPE also includes an initial estimate of the costs of dismantling and removing the item and restoring the site on which it is located, the obligation for which an entity incurs when the item is acquired. For example, a company may have rented an empty retail space in a local mall and proceeded to extensively renovate the space. It is very likely that the rental agreement will require the company to return the space to the landlord in its original state at the end of the rental period. Similarly, mining and oil exploration companies would typically have some form of environmental remediation obligation. The corresponding obligation component of the dismantling cost is accounted as a provision per *IAS 37—Provisions, Contingent Liabilities and Contingent Assets*. As this is a more advanced element of measurement of PPE on initial recognition, for our understanding, we can assume that there is no such dismantling obligations in subsequent discussions.

IAS 16 provides some examples of “directly attributable cost:”

- costs of employee benefits, like staff wages and salaries, arising directly from the construction or acquisition of the item of PPE
- costs of site preparation
- initial delivery and handling costs
- installation and assembly costs
- costs of testing whether the asset is functioning properly, after deducting the net proceeds from selling any items produced while bringing the asset to that location and condition (such as samples produced when testing equipment)
- professional fees

Similarly, *IAS 16* also provides examples of what should *not* be included in the cost of an item of PPE:

- costs of opening a new facility
- costs of introducing a new product or service (including costs of advertising and promotional activities)
- costs of conducting business in a new location or with a new class of customer (including costs of staff training)
- administration and other general overhead costs

Let’s apply this recognition and measurement criteria to a number of PPE items.

Land and Land Improvements

The cost of land includes its purchase price (cash plus any note payable given), brokerage commission, survey fees, legal fees, and any property taxes that the purchaser pays. Land cost also includes expenditures for grading and clearing the land and for removing unwanted buildings.

The cost of land does not include the cost of fencing, paving, security systems, lighting, and other similar items. These are separate PPE—called land improvements—and they are subject to depreciation. You saw earlier that Airbus’s PPE included a “land, leasehold improvements, and buildings” category. Whilst it is presented as a single category with a carrying amount of €5,169 million, Airbus’s accounts would have the specific breakdown of these items.

Suppose Airbus signs a €300,000 note payable to purchase a parcel of land for a new logistic site. Airbus also pays €10,000 for real estate commission, €13,000 of stamp duty, a €1,000 land survey fee, and €16,000 to pave the parking lot—all in cash. What should Airbus recognize as the cost of this land?

Purchase price of land	€300,000
Add related costs:	
Real estate commission	€10,000
Stamp duty	13,000
Survey fee	1,000
Total related costs	24,000
Total cost of land	<u>€324,000</u>

Note that the cost to pave the parking lot, €16,000, is *not* included in the land's cost, because the pavement is a land improvement. Airbus would record the purchase of this land as follows:

A1				
	A	B	C	D
1		Land	324,000	
2		Note Payable		300,000
3		Cash		24,000
4				

Assets	=	Liabilities	+	Shareholders' Equity
+ 324,000	=	+ 300,000	+	0
− 24,000				

This purchase of land increases both assets and liabilities. There is no effect on equity.¹

The cost to pave a parking lot (€16,000) would be recorded in a separate account entitled Land Improvements (or Leasehold Improvements if the land title is not in perpetuity). This account includes costs for such other items as driveways, signs, fences, and sprinkler systems. Although these assets are located on the land, they are subject to wear and tear and have a limited useful life, and their cost should therefore be depreciated over the term of the lease. Some companies call the depreciation on leasehold improvements *amortization*, which is the same concept as *depreciation*.

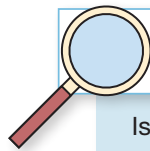
Buildings, Machinery, and Equipment

The cost of constructing a building includes architectural fees, building permits, contractors' charges, and payments for material, labor, and overhead. If the company constructs its own building, the cost will also include the cost of interest on money borrowed to finance the construction (if the recognition criteria in *IAS 23—Borrowing Costs* are met).

When an existing building (new or old) is purchased, its cost includes the purchase price, brokerage commission, sales and other taxes paid, and all expenditures to repair and renovate the building for its intended purpose.

The cost of Airbus's equipment includes its purchase price (less any discounts), plus transportation from the seller to Airbus, insurance while in transit, sales and other taxes, purchase commission, installation costs, and any expenditures to test the asset before it's placed in service. The equipment cost may also include the cost of any special platform necessary to place the equipment or other necessary safety measures. After the asset is up and running, insurance, taxes, and regular maintenance costs are recorded as expenses, not as part of the asset's cost.

¹ We show the accounting equation along with each journal entry—where the accounting equation aids your understanding of the transaction. Impact of revenue and expense transactions are taken directly to equity.



A Closer Look

Is a building sitting on a piece of land one asset or two? *IAS 16* suggests that these are separate assets that are to be accounted for separately, even when they are acquired together. Buildings have limited useful life and are always depreciated. On the other hand, land may be “freehold” (an estate owned for perpetuity) and not depreciated because it has an infinite useful life.

In many countries, however, land titles are not issued in perpetuity and usually have a limited tenure (for example, 30 years, 99 years, or even 999 years!), after which the title is returned to the leaseholder. This type of land is usually called “leasehold land.” Even for leasehold properties, the useful life of the building is usually not the same as the length of the leasehold tenure. That’s why we account for a building on a piece of land as two separate assets.

Lump-Sum (or Basket) Purchases of Assets

Businesses often purchase several assets as a group, or a “basket,” for a single lump-sum amount. For example, Airbus may pay one price for land and a building, but the company must first identify the cost of each asset. The total cost is then divided among the assets according to their relative sales (or market) values. This technique is called the *relative-sales-value method*.

Suppose Airbus purchases land and a building in Luxembourg. The building sits on two acres of land, and the combined purchase price of land and building is €2.7 million. An appraisal indicates that the land’s fair value is €1 million and the building’s fair value is €2 million. Airbus first figures the ratio of each asset’s fair value to the total fair value. The total appraised value is €1 + €2 million = €3 million. Thus, the land, valued at €1 million, is one-third of the total fair value. The building’s appraised value is two-thirds of the total. These percentages are then used to determine the cost of each asset, as follows:

Asset	Fair Value	Total Fair Value	Percentage of Total Fair Value	Total Cost	Cost of Each Asset
Land	€1,000,000	÷ €3,000,000	= 33.3%	× €2,700,000	€ 900,000
Building	2,000,000	÷ 3,000,000	= 66.6%	× €2,700,000	1,800,000
Total	€3,000,000		100%		€2,700,000

If Airbus pays cash, the entry to record the purchase of the land and building is

A1				
	A	B	C	D
1		Land	900,000	
2		Building	1,800,000	
3		Cash		2,700,000
4				

Assets	=	Liabilities	+	Shareholders' Equity
+ 900,000	=			
+ 1,800,000	=	0	+	0
– 2,700,000	=			

Total assets don’t change—only the makeup of Airbus’s assets will change.

» Stop & Think

How would Airbus divide a €120,000 lump-sum purchase price for land, building, and equipment with estimated market values of €40,000, €95,000, and €15,000, respectively?

Answer:

	Estimated Market Value	Percentage of Total Market Value	×	Total Cost	=	Cost of Each Asset
Land.....	€ 40,000	26.7%*	×	€120,000	=	€ 32,040
Building.....	95,000	63.3%	×	120,000	=	75,960
Equipment.....	15,000	10.0%	×	120,000	=	12,000
Total	<u>€150,000</u>	<u>100.0%</u>				<u>€120,000</u>

*€40,000/€150,000 = 0.267, and so on

UNDERSTAND WHEN TO CAPITALIZE OR EXPENSE SUBSEQUENT COSTS

Subsequent Costs

The PPE recognition criteria in *IAS 16* helps us in determining whether an expenditure should be recognized as an asset in the Balance Sheet or expensed immediately to the Income Statement. The same criteria also help us with expenditures subsequent to the initial recognition. Specifically, *IAS 16* states that an entity should *not* recognize the costs of the day-to-day servicing (which typically comprises the costs of labor and consumables, or small parts of the item) in the carrying amount of an item of PPE. These costs are expensed or charged to the Income Statement as incurred. The purpose of these expenditures is often described as for the “repairs and maintenance” of the PPE. For example, Airbus may perform regular maintenance of its motor vehicles. The costs of repainting an Airbus delivery truck, repairing its dented bumper, or replacing worn tires are also expensed immediately.

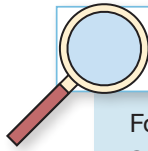
On the other hand, expenditures that increase the asset’s capacity or extend its useful life are called **capital expenditures**. For example, the cost of a major overhaul that extends the useful life of a Airbus truck is a capital expenditure. Capital expenditures are said to be “capitalized,” which means the cost is added to an asset account and not expensed immediately. Thus, a major decision in accounting for PPE is whether to capitalize or to expense a certain cost.

Continuing with our delivery truck example, Exhibit 7-4 shows the distinction between recognizing the capital expenditures as an asset and immediate charging the expenditure as an expense for the period.

Exhibit 7-4 Capital Expenditure or Immediate Expense for Costs Associated with a Delivery Truck

Record an Asset for Capital Expenditures	Record Repair and Maintenance Asset Expense
Significant or major repairs: Major engine overhaul Addition to storage capacity of truck Modification of body for new use of truck	Ordinary repairs: Repair of transmission or other mechanism Oil change, lubrication, and so on Replacement of tires and windshield, or a paint job

3 Understand
when to capitalize
or expense
subsequent costs



A Closer Look



For certain industries, it is possible that certain “repairs and maintenance” may be a necessary precondition to continue to operate the asset. For example, you would want to be sure that any airline you fly with has complied with all the required safety and maintenance checks. These are probably regular major inspections at certain points of the asset’s useful life or at preset usage intervals. *IAS 16* allows for the capitalization of these major inspections as part of the carrying amount of the item of property and allocated over the period (until the next inspection).

For example, Air France-KLM’s 2015 annual report states that: “Maintenance costs are recorded as expenses during the period when incurred, with the exception of programs that extend the useful life of the asset or increase its value, which are then capitalized (e.g., maintenance on airframes and engines, excluding parts with limited useful lives).” And similarly, Qantas’s 2016 annual report states: “The costs of subsequent major cyclical maintenance checks for owned and leased aircraft are recognized and depreciated over the shorter of the scheduled usage period to the next major inspection event or the remaining life of the aircraft or lease term (as appropriate) . . . All other maintenance costs are expensed as incurred.”

The distinction between a capital expenditure and an expense requires judgment: Does the cost extend the asset’s usefulness or its useful life? If so, record an asset. If the cost merely repairs the asset or returns it to its prior condition, then record an expense.

Most entities expense all expenditures below a certain threshold, say, \$1,000. Remember that there are always cost constraints in producing financial information. If the resulting information does not increase fundamental and enhancing qualitative characteristics, why incur unnecessary costs to produce the information? For higher costs, they follow the rule we gave earlier: capitalize costs that extend the asset’s usefulness or its useful life, and allocate the capitalized amount over the expected useful life of the asset.

Accounting errors sometimes occur for PPE costs. For example, a company may:

- Expense a cost that should have been capitalized. This error overstates expenses and understates net income in the year of the error.
- Capitalize a cost that should have been expensed. This error understates expenses and overstates net income in the year of the error.



COOKING THE BOOKS

by Improper Capitalization

WorldCom

It is one thing to accidentally capitalize an expense as PPE but quite another to do it intentionally, thus deliberately overstating assets, understating expenses, and overstating net income. One well-known company committed one of the biggest financial statement frauds in U.S. history in this way.

In 2002, WorldCom, Inc., was one of the largest telecommunications service providers in the world. The company had grown rapidly from a small, regional telephone company in 1983 to a giant corporation in 2002 by acquiring an ever-increasing number of other such companies. But 2002 was a bad year for WorldCom, as well as for many others in the “telecom” industry. The United States was reeling from the effects of a deep economic recession spawned by the “bursting dot-com bubble” in 2000 and intensified by the terrorist attacks on U.S. soil in 2001. Wall Street was looking high and low for positive signs, pressuring public companies to keep profits trending upward in order to support share prices, without much success, at least for the honest companies.

Bernard J. (“Bernie”) Ebbers, WorldCom’s chief executive officer, was worried. He began to press his chief financial officer, Scott Sullivan, to find a way to make the company’s Income

Statement look healthier. After all legitimate attempts to improve earnings failed, Sullivan concocted a scheme to cook the books.

Like all telecommunications companies, WorldCom had signed contracts with other telephone companies, paying them fees so that WorldCom customers could use their lines for telephone calls and Internet usage. Accounting standards require such fees to be expensed as incurred, rather than capitalized. Overestimating the growth of its business, WorldCom had incurred billions of dollars in such costs, about 15% more than its customers would ever use.

In direct violation of accounting standards, Sullivan rationalized that the excessive amounts WorldCom had spent on line costs would eventually lead to the company's recognizing revenue in future years (thus extending their usefulness and justifying, in his mind, their classification as assets). Sullivan directed the accountants working under him to reclassify line costs as property, plant, and equipment assets, rather than as expenses, and to amortize (spread) the costs over several years rather than to expense them in the periods in which they were incurred. Over several quarters, Sullivan and his assistants transferred a total of \$3.1 billion in such charges from operating expense accounts to PPE, resulting in the transformation of what would have been a net loss for all of 2001 and the first quarter of 2002 into a sizeable profit. It was the largest single fraud in U.S. history to that point.

Sullivan's fraudulent scheme was discovered by the company's internal audit staff during a routine spot-check of the company's records for capital expenditures. The staff members reported Sullivan's (and his staff's) fraudulent activities to the head of the company's audit committee and its external auditor, setting in motion a chain of events that resulted in Ebberts's and Sullivan's firing, and the company's eventual bankruptcy. Ebberts, Sullivan, and several of their assistants went to prison for their participation in this fraudulent scheme.

Shareholders of WorldCom lost billions of dollars in share value when the company went down, and more than 500,000 people lost their jobs. The WorldCom scandal rocked the financial world, causing global stock markets to plummet from lack of confidence. This scandal (as well as others such as Enron) eventually led to the passage of the U.S. Sarbanes-Oxley Act (see Chapter 5).

» Stop & Think

Ivana Low is the CFO of SMOO, an online retailing company. At the end of the fiscal year, the net profit before tax was \$100,000 (total revenue of \$300,000 and expenses of \$200,000). This was below the profit target set by the board of directors. Ivana noted that there was an expense of \$30,000 recorded at the end of the year for advertising during the year. Ivana decided to classify the expense as an asset with a useful life of 4 years in order to meet the profit target. What is the impact of this alternative accounting treatment on SMOO's pre-tax profit?

Answer:

The reclassification of an expense incurred at the end of the fiscal year as an asset will reduce expense, increase profit, and increase assets. The SMOO's pre-tax profit would now be (incorrectly) reported as: $\$300,000 - \$170,000 = \$130,000$.

MEASURE AND RECORD DEPRECIATION

As we've seen in previous chapters, PPE items are reported on the Balance Sheet at their carrying amounts or book values, which is:

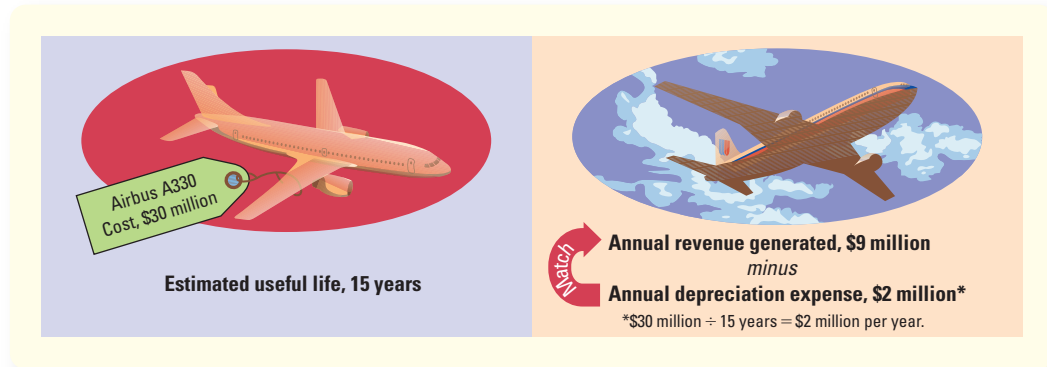
$$\text{Carrying amount of an item PPE} = \text{Cost} - \text{Accumulated Depreciation}$$

Depreciation is the systematic allocation of the cost of a long-term asset over its useful life. As the cost of an item of PPE is allocated to its useful lives, its carrying amounts are reduced and the

4 Measure and record depreciation

corresponding depreciation expense results in a reduction in equity. Exhibit 7-5 illustrates this concept with an example of an Airbus A330 owned by Brussels Airlines.

Exhibit 7-5 | Allocating Cost of Assets Over Useful Life



Recall that depreciation expense is charged periodically to the Income Statement. The cumulative amount of depreciation charged since the initial recognition and measurement of an asset is called **accumulated depreciation**, which you will find on the Balance Sheet (or in the notes to the accounts). At the start of this chapter, you saw the accumulated depreciation for all of Airbus's PPE (second column of lines 14–17).

You've just seen what depreciation is. Let's see what depreciation is *not*.

1. **Depreciation is not a process of valuation.** Businesses do *not* record depreciation based on changes in the market value of their PPEs.
2. **Depreciation does not mean setting aside cash to replace assets as they wear out.** Any cash fund set aside to purchase a new asset is entirely separate from depreciation.

How to Allocate Depreciation

Before we move to the specific depreciation methods, let's make sure we understand the basic concepts and terminologies related to depreciation. To allocate depreciation for a PPE item, we must know three things about the asset:

1. Cost
2. Estimated useful life
3. Estimated residual value

We have discussed cost earlier, which is a known amount. The other two factors must be estimated.

The economic benefit from owning a PPE is consumed by an entity primarily through the use of the PPE in the ordinary course of business over the PPE's useful life. As the allocation of expenses is usually predetermined in advance (we will discuss methods of depreciation in the next section), an entity such as Airbus would have to make an estimate of the useful lives of its buildings, leasehold improvements, plant and machinery, furniture and fixtures, office equipment, and motor vehicles. *IAS 16* indicates that the following factors ought to be considered in determining an asset's useful life:

- expected usage of the asset
- expected physical wear and tear, including the necessary repair and maintenance program
- technical or commercial obsolescence, including a change in the market demand for the product or service output of the asset

- legal or similar limits on the use of the asset, such as the expiry dates of Airbus's leasehold land

Estimated useful life is thus the length of service expected from using the asset. Useful life may be expressed in years, units of output, miles, or some other measure. For example, the useful life of Airbus's buildings, furniture and fixtures, and office equipment can be stated in years. Some of its specialized machinery may have its useful life expressed in number of units or usage or capacity. Occasionally, you may see this useful life expressed as a percentage, for example, a depreciation rate of 10% per year is equal to 10 years' useful life, 20% is equal to 5 years', etc. Companies base estimates on their experiences and trade publications. Useful life is not necessarily the same as the physical or economic life of the asset. An airplane may physically last over 30 years, but if the management's intention is to use it over a shorter time period, its useful life may be shorter. Singapore Airlines and Emirates Airline, for example, depreciate their passenger airplanes over 15 years' useful life, whereas Cathay Pacific and Thai Airways use 20 years' useful life, and British Airways uses 18–25 years.

Estimated residual value—also sometimes called *scrap value* or *salvage value*—of an asset is the estimated amount that an entity would currently obtain from disposal of the asset, after deducting the estimated costs of disposal, if the asset were already of the age and in the condition expected at the end of its useful life. For example, Airbus may believe that a package-handling machine will be useful for seven years. After that time, Airbus may expect to sell the machine as scrap metal. The amount Airbus believes it can get for the machine is the estimated residual value. In computing depreciation, the estimated residual value is not depreciated because Airbus expects to receive this amount from selling the asset. If there's no expected residual value, the full cost of the asset is depreciated. The residual value may be expressed as an absolute amount or as a percentage of the asset's cost. For example, Singapore Airlines uses a "10% residual value" in its depreciation allocation. A PPE's depreciable amount or **depreciable cost** is therefore:

$$\text{Depreciable Amount} = \text{Asset's cost} - \text{Estimated residual value}$$

Depreciation Methods

There are three main depreciation methods:

- constant allocation, i.e., straight-line
- by actual usages, i.e., units-of-production
- accelerated allocation, i.e., double-declining-balance

These methods allocate different amounts of depreciation to each period. However, they all result in the same total amount of depreciation, which is the asset's depreciable amount, over the life of the asset. Exhibit 7-6 presents the data we use to illustrate depreciation computations for an Airbus delivery truck.

Exhibit 7-6 | Data for Depreciation Computations: Example

Data Item	Amount
Cost of truck	€41,000
Less: Estimated residual value	(1,000)
Depreciable cost	<u>€40,000</u>
Estimated useful life:	
Years	5 years
Units of production	100,000 units [miles]



Straight-Line Method. In the **straight-line (SL) method**, an equal amount of depreciation is assigned to each year (or period) of asset use. Depreciable cost is divided by useful life in years to determine the annual depreciation expense. Applied to the truck data from Exhibit 7-6, SL depreciation is

$$\begin{aligned}\text{Straight-line depreciation per year} &= \frac{\text{Cost} - \text{Residual value}}{\text{Useful life, in years}} \\ &= \frac{€41,000 - €1,000}{5} \\ &= €8,000\end{aligned}$$

The entry to record depreciation is:

A1				
	A	B	C	D
1		Depreciation Expense	8,000	
2		Accumulated Depreciation		8,000
3				

Assets		=	Liabilities	+	Shareholders' Equity	
- 8,000		=	0	+	- 8,000	(depreciation expense)

Observe that depreciation decreases the asset (through Accumulated Depreciation) and also decreases equity (through Depreciation Expense). Let's assume that Airbus purchased this truck on January 1, 20X5. Exhibit 7-7 gives a *straight-line depreciation schedule* for the truck. The final column of the exhibit shows the asset's *carrying amount* or *net book value*, which is cost less accumulated depreciation. Note that we can develop this depreciation schedule before we have even bought the truck! Such schedules help us understand the impact of our purchase decisions on financial statements over the years.

Exhibit 7-7 Straight-Line Depreciation Schedule

A1							
	A	B	C			D	E
			Depreciation for the Year				
	Date	Asset Cost	Depreciation Rate*	Depreciable Amount	Depreciation Expense	Accumulated Depreciation	Asset Carrying Amount
1	1/1/20X5	€ 41,000					€ 41,000
2	12/31/20X5		0.20 ×	€40,000	= €8,000	€ 8,000	33,000
3	12/31/20X6		0.20 ×	40,000	= 8,000	16,000	25,000
4	12/31/20X7		0.20 ×	40,000	= 8,000	24,000	17,000
5	12/31/20X8		0.20 ×	40,000	= 8,000	32,000	9,000
6	12/31/20X9		0.20 ×	40,000	= 8,000	40,000	1,000
7							
8							
9							
10							
11							

*5 years' useful life = 20% (0.2) depreciation rate per year.

As an asset is used in operations:

- the accumulated depreciation increases.
- the carrying amount of the asset decreases.

An asset's final book value is its *residual value* (€1,000 in Exhibit 7-7). At the end of its useful life, the asset is said to be *fully depreciated*.

» Stop & Think

An item of Airbus office equipment that cost €10,000, has a useful life of three years, and has a residual value of €1,000 was purchased on January 1. What is its straight-line depreciation for each year?

Answer:

$$€3,000 = (€10,000 - €1,000)/3$$

Units-of-Production Method. In the **units-of-production (UOP) method**, a fixed amount of depreciation is assigned to each unit of output or service produced by the asset. Depreciable cost is divided by useful life—in units of production—to determine this amount. This per-unit depreciation expense is then multiplied by the number of units produced each period to compute depreciation. Obviously, the Dairy Farm delivery truck will not stop working just because it has been driven to 100,000 miles. We continue with the same delivery truck as an illustration, but in real life, this method is more likely to be used with assets with technical capacity or unit limitations rather than a delivery truck. The UOP depreciation for the Airbus truck data in Exhibit 7-6 is

$$\begin{aligned} \text{Units-of-production depreciation per unit of output} &= \frac{\text{Cost} - \text{Residual value}}{\text{Useful life, in units of production}} \\ &= \frac{€41,000 - €1,000}{100,000 \text{ miles}} = €0.40 \text{ per mile} \end{aligned}$$

Assume that Airbus expects to drive the truck 20,000 miles during the first year, 30,000 during the second, 25,000 during the third, 15,000 during the fourth, and 10,000 during the fifth. Exhibit 7-8 shows the UOP depreciation schedule.

Exhibit 7-8 Units-of-Production Depreciation Schedule

A1							
	A	B	C			D	E
1			Depreciation for the Year				
2							
3							
4	Date	Asset Cost	Depreciation Per Unit*	Number of Units	Depreciation Expense	Accumulated Depreciation	Asset Carrying Amount
5	1/1/20X5	€ 41,000					€ 41,000
6	12/31/20X5		€0.40* ×	€20,000 =	€ 8,000	€ 8,000	33,000
7	12/31/20X6		0.40 ×	30,000 =	12,000	20,000	21,000
8	12/31/20X7		0.40 ×	25,000 =	10,000	30,000	11,000
9	12/31/20X8		0.40 ×	15,000 =	6,000	36,000	5,000
10	12/31/20X9		0.40 ×	10,000 =	4,000	40,000	1,000
11							

*(€41,000 – €1,000)/100,000 miles = €0.40 per mile.

The amount of UOP depreciation varies with the number of units the asset produces in a year. In our example above, we have estimated the usage pattern, but the actual depreciation charge each year will be based on the actual outputs for the year. Thus, the amount charged each year may differ from what was originally planned.

For example, if the actual miles driven in 20X5 were 21,000, the depreciation charge for the year would have been €0.40 × 21,000 = €8,400. The UOP depreciation does not depend directly on passage of time, as do the other methods.

Double-Declining-Balance Method. An **accelerated depreciation method** (or using *IAS 16*'s terminology, “diminishing balance method”) writes off a larger amount of the asset's cost near the start of its useful life than the straight-line method does.

Double-declining-balance (DDB) is the main accelerated depreciation method and computes annual depreciation by multiplying the asset's declining book value by a constant percentage, which is double (or two times) the straight-line depreciation rate. DDB amounts are computed as follows:

- *First*, compute the straight-line depreciation rate per year. A five-year truck has a straight-line depreciation rate of $1/5$, or 20% each year. A 10-year asset has a straight-line rate of $1/10$, or 10%, and so on.
- *Second*, multiply the straight-line rate by 2 to compute the DDB rate. For a five-year asset, the DDB rate is 40% ($20\% \times 2$). A 10-year asset has a DDB rate of 20% ($10\% \times 2$). The DDB rate for the delivery truck in Exhibit 7-6 is:

$$\begin{aligned}\text{DDB depreciation rate per year} &= \frac{1}{\text{Useful life, in years}} \times 2 \\ &= \frac{1}{5 \text{ years}} \times 2 \\ &= 20\% \times 2 = 40\%\end{aligned}$$

- *Third*, multiply the DDB rate by the period's beginning asset book value (cost less accumulated depreciation). Under the DDB method, ignore the residual value of the asset in computing depreciation, except during the last year.
- *Fourth*, determine the final year's depreciation amount—that is, the amount needed to reduce asset book value to its residual value. In Exhibit 7-9, the fifth and final year's DDB depreciation is €4,314—book value of €5,314 less the €1,000 residual value. *The residual value should not be depreciated* but should remain on the books until the asset is disposed of.

Exhibit 7-9 Double-Declining-Balance Depreciation Schedule

A1							
	A	B	C			D	E
1	Depreciation for the Year						
2							
3		Asset	DDB	Asset Carrying	Depreciation	Accumulated	Asset
4	Date	Cost	Rate	Amount	Expense	Depreciation	Carrying
5	1/1/20X5	€ 41,000					€ 41,000
6	12/31/20X5		0.40 ×	€41,000 =	€16,400	€ 16,400	24,600
7	12/31/20X6		0.40 ×	24,600 =	9,840	26,240	14,760
8	12/31/20X7		0.40 ×	14,760 =	5,904	32,144	8,856
9	12/31/20X8		0.40 ×	8,856 =	3,542	35,686	5,314
10	12/31/20X9				4,314*	40,000	1,000
11							

*Last-year depreciation is the “plug” amount needed to reduce asset book value (far right column) to the residual amount (€5,314 – €1,000 = €4,314).

The DDB method differs from the other methods in three ways:

1. It is an accelerated depreciation method, so depreciation expenses in the early years are significantly more than in later years.
2. Residual value is ignored initially; first-year depreciation is computed on the asset's full cost.
3. Depreciation expense in the final year is the “plug” amount needed to reduce the asset's book value to the residual amount.

» Stop & Think

What is the DDB depreciation each year for the asset in the Stop & Think on page 401?

Answers:

Yr. 1: €4,000 ($€10,000 \times 40\%$)
 Yr. 2: €2,400 ($€6,000 \times 40\%$)
 Yr. 3: €1,440 ($€3,600 \times 40\%$)
 Yr. 4: €160 ($€10,000 - €4,000 - €2,400 - €1,440 - €2,000 = €160$)*
 Yr. 5: €0

*The asset is not depreciated below residual value of €2,000.

Comparing Depreciation Methods

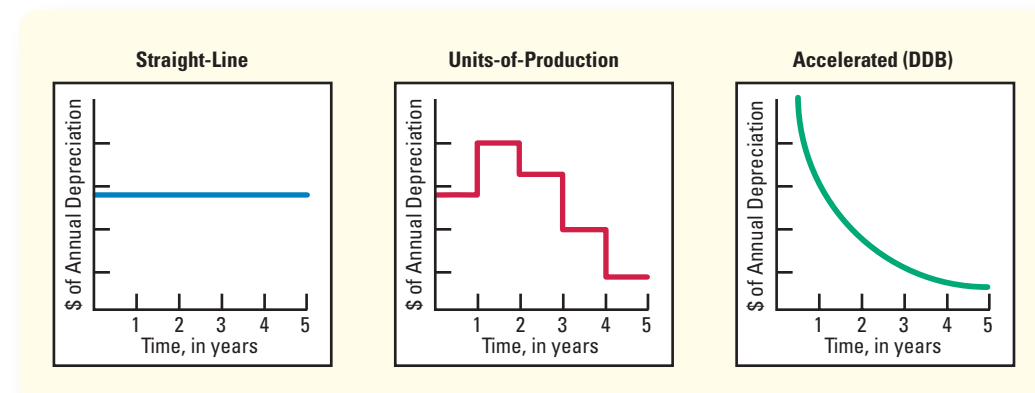
Let's compare the three methods in terms of the yearly amount of depreciation. The yearly amount varies by method, but the total €40,000 depreciable cost, i.e., the total accumulated depreciation at the end of the asset's life, is the same under all methods.

Amount of Depreciation Expense per Year			
Year	Straight-Line	Estimated Units-of-Production	Accelerated Method Double-Declining Balance
1	€ 8,000	€ 8,000	€16,400
2	8,000	12,000	9,840
3	8,000	10,000	5,904
4	8,000	6,000	3,542
5	8,000	4,000	4,314
Total	<u>€40,000</u>	<u>€40,000</u>	<u>€40,000</u>

Exhibit 7-10 graphs annual depreciation amounts for the straight-line, units-of-production, and accelerated depreciation (DDB) methods. The graph of straight-line depreciation is flat through time because annual depreciation is the same in all periods.

Units-of-production depreciation follows no particular pattern because annual depreciation depends on the actual use of the asset during the year. Accelerated depreciation is greatest in the first year and less in the later years.

Exhibit 7-10 Depreciation Patterns Through Time



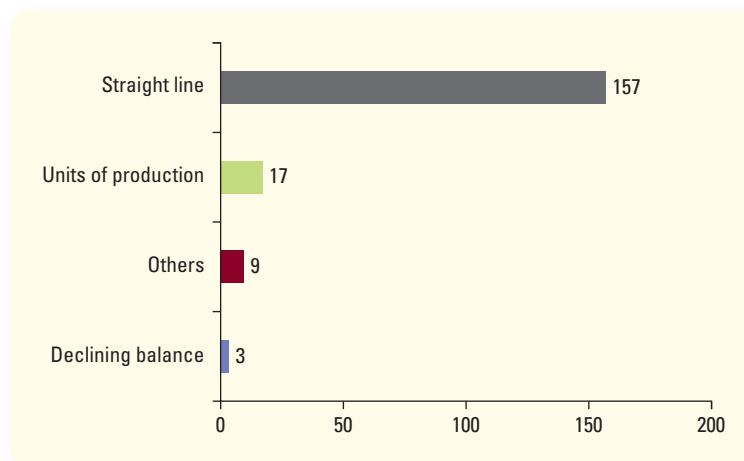
Choosing a Depreciation Method

Clearly, the choice of a depreciation method will impact the profit of any entity. How should an entity choose the “right” depreciation method? *IAS 16* requires that the depreciation method chosen ought to reflect the *pattern of consumption* of the economic benefits embodied in the asset. At every financial year-end, an entity should review the depreciation method, and unless there is a significant change in the pattern of consumption, it should continue to apply the method consistently from period to period.

For PPE assets with a reasonably constant pattern of consumption, the allocation basis should be done using the straight-line method. The units-of-production method best fits those assets that wear out because of physical use rather than obsolescence.

An accelerated method (such as DDB) applies best to assets that generate more revenue earlier in their useful lives and less in later years. Exhibit 7-11 shows a recent study of 170 IFRS companies and their depreciation method (AICPA 2010). It is clear that a very significant majority of the survey companies use the straight-line method of depreciation. About 20 companies in the survey use more than one method of depreciation.

Exhibit 7-11 Depreciation Methods Used by 170 Companies



Airbus uses the following useful lives for its assets, and the primary method used for depreciation is the straight-line method.

ADAPTED EXCERPTS FROM AIRBUS GROUP'S NOTES TO THE ACCOUNTS

Property, Plant, and Equipment

Items of PPE are generally depreciated on a straight-line basis. The following useful lives are assumed:

Buildings	10 to 50 years
Site improvements	6 to 30 years
Technical equipment and machinery	3 to 20 years
Jigs and tools (1)	5 years
Other equipment, factory and office equipment	2 to 10 years

(1) If more appropriate, jigs and tools are depreciated using the number of production or similar units expected to be obtained from the tools (units-of-production method).

Source: Airbus Group's 2015 Financial Statements, page 36

Mid-Chapter

Summary Problem

Suppose Airbus purchased equipment on January 1, 20X6, for €44,000. The expected useful life of the equipment is 10 years or 100,000 units of production, and its residual value is €4,000. Under three depreciation methods, the annual depreciation expense and the balance of accumulated depreciation at the end of 20X6 and 20X7 are as follows:

Year	Method A		Method B		Method C	
	Annual Depreciation Expense	Accumulated Depreciation	Annual Depreciation Expense	Accumulated Depreciation	Annual Depreciation Expense	Accumulated Depreciation
20X6	€4,000	€4,000	€8,800	€ 8,800	€1,200	€1,200
20X7	4,000	8,000	7,040	15,840	5,600	6,800

Requirements

1. Identify the depreciation method used in each instance, and show the equation and computation for each. (Round to the nearest euro.)
2. Assume continued use of the same method through year 20X8. Determine the annual depreciation expense, accumulated depreciation, and carrying amount (or book value) of the equipment for 20X6 through 20X8 under each method, assuming 12,000 units of production in 20X8.

Answers

Requirement 1

Method A: Straight-Line

$$\text{Depreciable amount} = €40,000 (€44,000 - €4,000)$$

$$\text{Each year: } €40,000 / 10 \text{ years} = €4,000$$

Method B: Double-Declining-Balance

$$\text{Rate} = \frac{1}{10 \text{ years}} \times 2 = 10\% \times 2 = 20\%$$

$$20X6: 0.20 \times €44,000 = €8,800$$

$$20X7: 0.20 \times (€44,000 - €8,800) = €7,040$$

Method C: Units-of-Production

$$\text{Depreciation per unit} = \frac{€44,000 - €4,000}{100,000 \text{ units}} = €0.40$$

$$20X6: €0.40 \times 3,000 \text{ units} = €1,200$$

$$20X7: €0.40 \times 14,000 \text{ units} = €5,600$$

Requirement 2

Method A: Straight-Line			
Year	Annual Depreciation Expense	Accumulated Depreciation	Book Value
Start			€44,000
20X6	€4,000	€ 4,000	40,000
20X7	4,000	8,000	36,000
20X8	4,000	12,000	32,000

Method B: Double-Declining-Balance			
Year	Annual Depreciation Expense	Accumulated Depreciation	Book Value
Start			€44,000
20X6	€8,800	€ 8,800	35,200
20X7	7,040	15,840	28,160
20X8	5,632	21,472	22,528

Method C: Units-of-Production			
Year	Annual Depreciation Expense	Accumulated Depreciation	Book Value
Start			€44,000
20X6	€1,200	€ 1,200	42,800
20X7	5,600	6,800	37,200
20X8	4,800	11,600	32,400

Computations for 20X8			
Straight-line	$€40,000/10 \text{ years} = €4,000$		
Double-declining-balance	$€28,160 \times 0.20 = €5,632$		
Units-of-production	$12,000 \text{ units} \times €0.40 = €4,800$		

Other Issues in Accounting for PPE

Accounting for PPE may also need to handle additional issues related to:

- choice of depreciation method, which may affect income taxes; a different depreciation method may be used for financial reporting versus tax purposes
- depreciation for partial years
- the fact that PPE have long lives; subsequently, better information may change estimates of useful life of assets and residual values
- alternative models for measurement of PPE subsequent to initial recognition
- companies that have gains or losses when they sell PPE

Let's take a brief look at some of these issues.

Depreciation for Tax Purposes

Airbus and most other companies use straight-line depreciation for reporting to shareholders and creditors on their financial statements. But for tax purposes, they may keep a separate set of depreciation records, depending on the specific tax regulations in various jurisdictions. There are two primary reasons why this is the case.

First, certain jurisdictions may mandate a specific treatment for specific assets. For example, in Singapore, the depreciation of Airbus's commercial vehicles, such as its delivery trucks, may be used to claim capital allowances (i.e., deductions from taxable income), but the depreciation of non-commercial vehicles, such as motor vehicles for its senior management staff, would not be allowed as a deduction. In other countries, there may be a maximum cap on the depreciable amount allowed for certain assets. Clearly, in order to comply with taxation rules, a different set of depreciation records will be required.

Second, tax regulations could provide alternative depreciation methods or schedules that are more favorable than what is being used for financial reporting. For example, investment in “green technologies” may be granted a double tax deduction to encourage businesses to make headway in the fight against climate change. You may have spent \$1 million on such equipment, but you are entitled to claim \$2 million as deductions in your tax forms. In other instances, you may be using the straight-line method for financial reporting, but the tax regulations may allow you to use an accelerated method such as DDB for tax reporting.

Suppose you are an Airbus country manager, and your local tax authority allows an accelerated depreciation method. Why would you prefer accelerated over straight-line depreciation for income tax purposes? This choice is easy. Accelerated depreciation provides the fastest tax deductions, thus decreasing immediate tax payments. Airbus can reinvest the tax savings back into the business or pay off interest-bearing debts (see Exhibit 7-12).

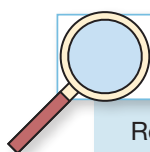
To understand the relationships between cash flow, depreciation, and income tax, recall our depreciation example of an Airbus delivery truck:

- First-year depreciation is €8,000 under straight-line and €16,400 under double-declining balance (DDB).
- DDB is permitted for income tax purposes, and headline corporate tax rate is 30%.

Exhibit 7-12 | The Cash-Flow Advantage of Accelerated Depreciation over Straight-Line Depreciation for Income Tax Purposes

	SL	Accelerated
1 Cash revenue.....	€400,000	€400,000
2 Cash operating expenses	300,000	300,000
3 Cash provided by operations before income tax.....	100,000	100,000
4 Depreciation expense (a noncash expense).....	8,000	16,400
5 Income before income tax	€ 92,000	€ 83,600
6 Income tax expense (30%)	€ 27,600	€ 25,080
Cash-flow analysis:		
7 Cash provided by operations before tax	€100,000	€100,000
8 Income tax expense	27,600	25,080
9 Cash provided by operations	€ 72,400	€ 74,920
10 Extra cash available for investment or debt repayment if DDB is used (€74,920 – €72,400).....		€ 2,520

You can see that, for income tax purposes, accelerated depreciation helps conserve cash for the business. That’s why virtually all companies will choose accelerated depreciation to compute their income tax, if allowed.



A Closer Look

Remember that there are two different “profits”: one is the net profit in your financial statements, which are prepared in accordance with the applicable financial standards, and the other is taxable income on your tax filings, which are prepared in accordance with the applicable taxation rules. To understand more about taxation for accounting purposes, you may cover *IAS 12—Income Taxes* in your future courses, but if your interest is on the tax reporting side, you will need to read the various taxation rules and regulations. *IAS 12* is beyond our course coverage, but it results in

some accounts that you may encounter in many companies: *deferred tax assets* and *deferred tax liabilities*. These accounts arise due to the difference between net profit (in financial reporting) and taxable income (in tax reporting). This does not mean that we are “keeping two sets of books,” which is usually associated with unscrupulous behavior and cheating. We need to maintain two separate schedules because the rules are different, not because we are trying to hide income from the tax authorities!

Depreciation for Partial Years

Companies purchase PPE whenever they need them, not just at the beginning of the year. Therefore, companies must compute depreciation for partial years or whenever they need to report to shareholders (e.g., quarterly or half-yearly reports). Suppose Airbus purchases a warehouse building on April 1 for €500,000. The building’s estimated life is 20 years, and its estimated residual value is €80,000. Airbus’s fiscal year-end is December 31. Let’s consider how Airbus computes depreciation for April through December:

- First, compute depreciation for a full year.
- Second, multiply full-year depreciation by the fraction of the year that you held the asset—in this case, 9/12. Assuming the straight-line method, the year’s depreciation for this Airbus building is €15,750, as follows:

$$\begin{array}{ll} \text{Full-year depreciation} & \frac{€500,000 - €80,000}{20} = €21,000 \\ \text{Partial year depreciation} & €21,000 \times 9/12 = €15,750 \end{array}$$

What if Airbus bought the asset on April 18? Many businesses record no monthly depreciation on assets purchased after the 15th of the month, and they record a full month’s depreciation on an asset bought on or before the 15th. Actual practices may vary from company to company, but in the overall scheme of things, a difference of a few days is not likely to be material for long-lived assets. Most companies use computerized systems to account for fixed assets. Each depreciable asset has a unique identification number, and the system will automatically calculate the asset’s depreciation expense. Accumulated Depreciation will then be automatically updated.

Changes in Estimates of Useful Lives or Residual Values

After an asset is in use, managers may change its useful life on the basis of experience and new information. This is not something that would happen often. Here’s an example from Lenovo in 2008:



EXCERPTS (ADAPTED) FROM LENOVO'S 2008 NOTES TO THE FINANCIAL STATEMENTS

During the year, the estimated useful life of tooling equipment was reviewed and changed from 10–20 years to two years as it reflects the current product life cycle. This change has resulted in an accelerated depreciation charge of approximately \$37 million.

© 2008 Lenovo

As you can see from the disclosure above, the change to a shorter useful life increased the depreciation charge for the year. This is called a change in accounting estimates (*IAS 8—Accounting Policies, Changes in Accounting Estimates and Errors*). Changes in estimates are accounted for *prospectively*, which means “from now on.” Lenovo recalculated depreciation on the basis of revised useful lives of its tooling equipment. Changes in estimates may also occur for residual values and are accounted for similarly.

Assume that Lenovo bought equipment costing \$50,000 and that the company originally believed the asset had a 10-year useful life with no residual value. Using the straight-line method, the company would record \$5,000 depreciation each year (\$50,000/10 years = \$5,000). Suppose Lenovo used the asset for four years. Accumulated depreciation reached \$20,000, leaving a remaining depreciable book value (cost less accumulated depreciation less residual value) of \$30,000 (\$50,000 – \$20,000). Based on the asset's conditions at the end of year four, management believes the asset will remain useful for eight more years. The company would spread the remaining depreciable book value over the asset's remaining life as follows:

$$\begin{array}{rcl} \text{Asset's remaining} & \div & (\text{New}) \text{ Estimated} \\ \text{depreciable book value} & & \text{useful life remaining} \\ \$30,000 & \div & 8 \text{ years} \\ & & = \\ & & \$3,750 \end{array} = \begin{array}{rcl} (\text{New}) \text{ Annual} \\ \text{depreciation} \end{array}$$

The yearly depreciation entry based on the new estimated useful life is

A1				
	A	B	C	D
1		Depreciation Expense—Equipment	3,750	
2		Accumulated Depreciation—Equipment		3,750
3				

Depreciation decreases both assets and equity.

Assets	=	Liabilities	+	Shareholders' Equity	
– 3,750	=	0	+	– 3,750	(depreciation expense)



COOKING THE BOOKS

Through Depreciation

Waste Management

Since PPEs usually involve relatively large amounts and relatively large numbers of assets, sometimes a seemingly subtle change in the way they are accounted for can have a tremendous impact on the financial statements. When these changes are made in order to cook the books, the results can be devastating.

Waste Management, Inc., is North America's largest integrated waste service company, providing collection, transfer, recycling, disposal, and waste-to-energy services for commercial, industrial, municipal, and residential customers from coast to coast.

Starting in 1992, six top executives of the company, including its founder and chairman of the board, its chief financial officer, its corporate controller, its top lawyer, and its vice president of finance, decided that the company's profits were not growing fast enough to meet "earnings targets," which were tied to their executive bonuses. Among several fraudulent financial tactics these top executives employed to cook the books were: (1) assigning unsupported and inflated salvage values to garbage trucks; (2) unjustifiably extending the estimated useful lives of their garbage trucks; and (3) assigning arbitrary salvage values to other fixed assets that previously had no salvage values. All of these tactics had the effect of decreasing the amount of depreciation expense in the Income Statements and increasing net income by a corresponding amount. While practices like this might seem relatively subtle and even insignificant when performed on an individual asset, remember that there were thousands of trash trucks and dumpsters involved, so the dollar amount grew huge in a short time. In addition, the company continued these practices for five years, overstating earnings by \$1.7 billion.

The Waste Management fraud was the largest of its kind in history until the WorldCom scandal, discussed earlier in this chapter. In 1997, the company fired the officers involved and hired a new CEO who ordered a review of these practices, which uncovered the fraud. In the

meantime, these dishonest executives had profited handsomely, receiving performance-based bonuses based on the company's inflated earnings, retaining their high-paying jobs, and receiving enhanced retirement benefits. One of the executives took the fraud to another level. Just 10 days before the fraud was disclosed, he enriched himself with a tax benefit by donating inflated company shares to his alma mater to fund a building in his name! Although the men involved were sued for monetary damages, none of them ever went to jail.

When the fraud was disclosed, Waste Management shareholders lost over \$6 billion in the market value of their investments when the share price plummeted by more than 33%. The company and these officers eventually settled civil lawsuits for approximately \$700 million because of the fraud.

You might ask, "Where were the auditors while this was occurring?" The company's auditor was Arthur Andersen, LLP, whose partners involved on the audit engagement were eventually found to be complicit in the scheme. In fact, a few of the Waste Management officers who perpetrated the scheme had been ex-partners of the audit firm. As it turns out, the auditors actually identified many of the improper accounting practices of Waste Management. However, rather than insisting that the company fix the errors, or risk exposure, they merely "persuaded" management to agree not to repeat these practices in the future, and entered into an agreement with them to write off the accumulated Balance Sheet overstatement over a period of 10 years. In June 2001, the SEC fined Arthur Andersen \$7 million for "knowingly and recklessly issuing false and misleading audit reports" for Waste Management from 1993 through 1996.

In October 2001, immediately on the heels of these disclosures, the notorious Enron scandal broke. Enron, as well as WorldCom, were Arthur Andersen clients at the time. The Enron scandal (discussed in Chapter 10) finally put the firm out of business. Many people felt that, had it not been for Andersen's involvement in the Waste Management affair, the SEC might have been more lenient toward the company in the Enron scandal.

Impairment of PPE

As you probably know, the fight for the next generation format was won by Sony's Blu-ray when Toshiba abandoned the HD DVD format in February 2008. Prior to this, Toshiba had experienced significant difficulties over a period of time. Warner Brothers, Wal-Mart, Best-Buy, and many others had started to stop the sales of HD DVD, causing a severe drop in demand for it. Suppose that at the start of the format war, Toshiba had a dedicated factory costing \$1 billion that produced HD DVD, which was being depreciated over its estimated useful life of 10 years on a straight-line basis. After three years, the carrying amount of the equipment would have been \$700 million. In this scenario, should Toshiba have continued depreciating the factory over 10 years, in light of the significant changes in the market for the outputs of its factory?

This is an example of how an asset may be impaired. *IAS 36—Impairment of Assets* provides guidance on this matter. An asset is impaired when its carrying amount is higher than its recoverable amount. **Recoverable amount** is the higher of fair value less cost to sell and value-in-use. The exact determination of the recoverable amount, and many other aspects of impairment of assets, is beyond an introductory accounting course. However, it is important for you to know the basic concepts of impairment. Many companies in the financial crisis have reported billions of impairment losses.

Suppose that when the carrying amount of the factory was \$700 million, the fair value less cost to sell was \$300 million and the value-in-use was \$100 million. First, we determine that the recoverable amount was \$300 million (higher of the two amounts). Toshiba would then recognize an impairment loss of \$400 million (from \$700 to \$300 million) with the following journal entry (in millions):

A1				
	A	B	C	D
1	Year 3	Impairment Loss on Factory (\$700 – \$300)	400	
2		Accumulated Depreciation and Impairment Loss		400
3				

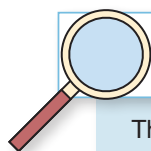
Both assets (Factory) and equity decrease (through the Loss account). Under IFRS, reversal of impairment losses may be permitted under certain limited circumstances.

Assets	=	Liabilities	+	Shareholders' Equity	
- 400	=	0	+	- 400	(impairment loss)

Measurement Subsequent to Initial Recognition

Under *IAS 16*, an entity elects one out of two measurement models for each class of property, which is defined as a grouping of assets of similar nature and use in an entity's operations. For example, Airbus uses five classes of PPE: Freehold properties, Leasehold properties, Leasehold improvements, Plant and machinery, and Furniture, equipment and motor vehicles.

- **Cost model:** An item of PPE shall be carried at its cost, less any accumulated depreciation and any accumulated impairment losses. This is similar to what we have discussed in this chapter thus far.
- **Revaluation model:** An item of PPE whose fair value can be measured reliably shall be carried at a revalued amount, being its fair value at the date of the revaluation less any subsequent accumulated depreciation and subsequent accumulated impairment losses. Revaluations shall be made with sufficient regularity to ensure that the carrying amount does not differ materially from that which would be determined using fair value at the Balance Sheet dates.



A Closer Look

The revaluation model is a little more complicated than the cost model. *IAS 16* provides additional guidelines on the determination of fair values, the frequency of revaluations, the treatment of revaluation gains and losses, and adjustments to accumulated depreciation. Your instructor may refer you to *IAS 16* if additional coverage of the operations of revaluation model is required for your course.

Suppose you have a PPE item with a carrying amount of \$100,000 (cost of \$150,000 less \$50,000 accumulated depreciation) and elected to use the revaluation model for this class of PPE. The fair value amount is reliably determined to be \$120,000. One common way to handle the revaluation is to "restate" the asset at the new amount with zero depreciation. Using this approach, the following journal entry is entered.

A1				
	A	B	C	D
1	Jan. XX	Accumulated depreciation	50,000	
2	20X1	PPE revalued (150,000 – 120,000)		30,000
3		Revaluation adjustment		20,000
4		Revalued PPE from \$150,000–\$50,000 to \$120,000		
5				

After posting this entry, the accumulated depreciation is now zero, and the PPE is carried at the new fair value of \$120,000. The revaluation adjustment is an equity account and will be shown as other comprehensive income in the Statement of Comprehensive Income.

Using Fully Depreciated Assets

A *fully depreciated asset* is one that has reached the end of its estimated useful life. Suppose Airbus has fully depreciated equipment with zero residual value (cost was \$60,000). Airbus's accounts will appear as follows:

Equipment	–	Accumulated Depreciation	=	Book value
60,000		60,000		\$0

The equipment's book value is zero, but that doesn't mean the equipment is worthless. Airbus may use the equipment for a few more years, but Airbus will not record any more depreciation on a fully depreciated asset.

When Airbus disposes of the equipment, Airbus will remove both the asset's cost (\$60,000) and its accumulated depreciation (\$60,000) from the books. The next section shows how PPE disposals are accounted.

ACCOUNT FOR PPE DISPOSALS

5 Analyze the effects of a PPE disposal

Eventually, a PPE will cease to serve a company's needs. The asset may wear out or become obsolete. Before accounting for the disposal of the asset, the business should bring depreciation up-to-date to:

- Update the asset's final book value.
- Record the expense up to the date of disposal.

To account for disposal, remove the asset and its related accumulated depreciation from the books. Suppose the final year's depreciation expense has just been recorded for a machine that cost \$60,000 and is estimated to have zero residual value. The machine's accumulated depreciation thus totals \$60,000. Assuming that this asset is junked or scrapped, the entry to record its disposal is:

A1				
	A	B	C	D
1		Accumulated Depreciation—Machinery	60,000	
2		Machinery		60,000
3		<i>To dispose of a fully depreciated machine.</i>		
4				

Assets	=	Liabilities	+	Shareholders' Equity
+ 60,000	=	0	+	0
– 60,000				

There is no gain or loss on this disposal, and there's no effect on total assets, liabilities, or equity. If assets are junked, scrapped, or disposed of before being fully depreciated, the company incurs a loss on the disposal. Suppose the company disposes of equipment that cost \$60,000. This asset's accumulated depreciation is \$50,000, and its book value is, therefore, \$10,000. Scrapping this equipment results in a loss equal to the carrying amount of the asset, as follows:

A1				
	A	B	C	D
1		Accumulated Depreciation—Equipment	50,000	
2		Loss on Disposal of Equipment	10,000	
3		Equipment		60,000
4		<i>To dispose of equipment.</i>		
5				

Assets	=	Liabilities	+	Shareholders' Equity
+ 50,000	=	0	+	- 10,000 (loss on disposal of PPE)
- 60,000				

The company disposed of an asset with a \$10,000 book value and received nothing. The result is a \$10,000 loss, which decreases both total assets and equity. The (gain) loss on disposal of equipment is typically reported as other income (expense) on the Income Statement. Losses decrease net income exactly as expenses do. Gains increase net income in the same way as revenues.

Selling a PPE. Suppose Zar Zar Trading sells equipment on September 30, 20X8, for \$7,300 cash. The equipment cost \$10,000 when purchased on January 1, 20X5, and has been depreciated straight-line. Zar Zar estimated a 10-year useful life and no residual value. Prior to recording the sale, Zar Zar's accountants must update the asset's depreciation. Partial-year depreciation must be recorded for the asset's depreciation from January 1, 20X8, to the sale date. The straight-line depreciation entry at September 30, 20X8, is

A1				
	A	B	C	D
1	Sept. 30	Depreciation Expense (\$10,000/10 years × 9/12)	750	
2		Accumulated Depreciation—Equipment		750
3		To update depreciation.		
4				

The Equipment account and the Accumulated Depreciation account appear as follows. Observe that the equipment's book value is \$6,250 (\$10,000 – \$3,750).

Equipment		Accumulated Depreciation		
Jan. 1, 20X5	10,000		Dec. 31, 20X5	1,000
			Dec. 31, 20X6	1,000
			Dec. 31, 20X7	1,000
			Sept. 30, 20X8	750
			Balance	3,750

= Book value \$6,250

The gain on the sale of the equipment for \$7,300 is \$1,050, computed as follows:

Cash received from sale of the asset	\$7,300
Book value of asset sold:	
Cost	\$10,000
Less: Accumulated depreciation	(3,750)
Gain on sale of the asset	<u>\$1,050</u>

The entry to record sale of the equipment is:

A1				
	A	B	C	D
1	Sept. 30	Cash	7,300	
2		Accumulated Depreciation—Equipment	3,750	
3		Equipment		10,000
4		Gain on Sale of Equipment		1,050
5		To sell equipment.		
6				

This shows that the total assets will increase along with equity—by the amount of the gain.

Assets	=	Liabilities	+	Shareholders' Equity
+ 7,300				
+ 3,750	=	0	+	1,050 (gain from sale of equipment)
– 10,000				

Gains are recorded as credits. Gains and losses on asset disposals appear on the Income Statement as Other income (expense) or Other gains (losses). What if Zar Zar sold the same asset on September 30, 20X8, for \$5,000? Recall that the book value on the date of the sale was \$6,250. Thus, a loss of \$1,250 on disposal of PPE would be recognized.

A1				
	A	B	C	D
1	Sept. 30	Cash	5,000	
2		Accumulated Depreciation—Equipment	3,750	
3		Loss on Sale of PPE	1,250	
4		Equipment		10,000
5		<i>To sell equipment.</i>		
6				

Total assets decrease, and equity decreases—by the amount of the loss.

Assets	=	Liabilities	+	Shareholders' Equity
+ 5,000				
+ 3,750	=	0	+	– 1,250 (loss on sale of equipment)
– 10,000				

Exchanging PPE. Managers often trade in old assets for new ones. This is called a *non-monetary exchange*. The accounting for non-monetary exchanges is based on the *fair values of the assets involved*. Thus, the cost of an asset like plant and equipment received in a non-monetary exchange is equal to the fair values of the assets given up (including the old asset and any cash paid). Any difference between the fair value of the old asset from its book value is recognized as a gain (fair value of old asset exceeds book value) or a loss (book value of old asset exceeds fair value) on the exchange.

For example, assume Zar Zar has an old delivery car that cost \$9,000 and has accumulated depreciation of \$8,000. Thus, the old car's book value is \$1,000. Zar Zar trades in the old automobile for a new one with a fair market value of \$15,000 and pays cash of \$10,000. Thus, the implied fair value of the old car is \$5,000 (\$15,000 – \$10,000). This amount is treated as cash paid by the seller for the old vehicle. The cost of the new delivery car is \$15,000 (fair value of the old asset, \$5,000, plus cash paid, \$10,000). Zar Zar would record the exchange transaction as follows:

A1				
	A	B	C	D
1		Delivery Auto (new)	15,000	
2		Accumulated Depreciation (old)	8,000	
3		Delivery Auto (old)		9,000
4		Cash		10,000
5		Gain on Exchange of Delivery Auto		4,000
6		<i>Traded in old delivery car for new auto.</i>		
7				

Assets	=	Liabilities	+	Shareholders' Equity	
+ 15,000					
+ 8,000	=	0	+	4,000	(gain on exchange of assets)
- 9,000					
- 10,000					

There was a net increase in total assets of \$4,000 and a corresponding increase in shareholders' equity, to reflect the gain on the exchange. Notice that this amount represents the excess of the fair value of the old asset over its book value. Some other special rules may apply here, but they are covered in more advanced courses.

T-Accounts for Analyzing PPE Transactions

You can perform quite a bit of analysis if you know how transactions affect the PPE accounts. Here are the accounts with descriptions of the activity in each account.

PPE		Accumulated Depreciation	
Beg. bal.		Acc. depn. of assets disposed of	Beg. bal.
Cost of assets purchased	Cost of assets disposed of		Depreciation expense for the current period
End. bal.			End. bal.

Depreciation Expense		Gain on Sale of PPE	
Depreciation expense for the current period			Gain on sale
		OR	
		Loss on Sale of PPE	
		Loss on sale	

Example: Suppose you started the year with buildings that cost \$100,000. During the year you bought another building for \$150,000 and ended the year with buildings that cost \$180,000. What was the cost of the building you sold?

Building			
Beg. bal.	100,000	Cost of assets sold	?
Cost of assets purchased	150,000		= \$70,000*
End. bal.	180,000		

*100,000 + 150,000 - \$180,000

You can perform similar analyses to answer other interesting questions about what the business did during the period.

Accounting for Natural Resources

Natural resources are PPE of a special type, such as iron ore, petroleum (oil), and timber. As PPE are expensed through depreciation, natural resource assets are expensed through *depletion*. **Depletion expense** is that portion of the cost of a natural resource that is used up in a particular period. Depletion expense is computed in the same way as units-of-production depreciation.

An oil lease may cost Royal Dutch Shell \$100,000 and contain an estimated 10,000 barrels of oil. The depletion rate would be \$10 per barrel (\$100,000/10,000 barrels). If 3,000 barrels are extracted, depletion expense is \$30,000 (3,000 barrels × \$10 per barrel). The depletion entry is:

A1				
	A	B	C	D
1		Depletion Expense (3,000 barrels × \$10)	30,000	
2		Accumulated Depletion—Oil		30,000
3				

This entry is almost identical to a depreciation entry using the units-of-production method. If 4,500 barrels are removed the next year, that period's depletion is \$45,000 (4,500 barrels $\times$ \$10 per barrel). Accumulated Depletion is a contra account similar to Accumulated Depreciation.

Natural resource assets can be reported on Shell's Balance Sheet as follows (amounts assumed):

Property, Plant, and Equipment:		
Equipment.....	\$960,000	
Less: Accumulated depreciation	<u>(410,000)</u>	\$550,000
Oil.....	\$340,000	
Less: Accumulated depletion	<u>(140,000)</u>	200,000
Total property, plant, and equipment.....		<u>\$750,000</u>

6 Understand the recognition and subsequent measurement of intangible assets

UNDERSTAND THE RECOGNITION AND SUBSEQUENT MEASUREMENT OF INTANGIBLE ASSETS

As we have discussed at the start of this chapter, *intangible assets* are identifiable, long-lived assets without physical substance. Intangibles are valuable because they carry special rights from patents, copyrights, trademarks, franchises, leaseholds, and goodwill. If you look back at Airbus's Balance Sheet at the start of this chapter, you may notice that its intangible assets (line 6) amounted to €12,555 million, about 12% of total non-current assets. Intangibles are the most valuable assets for high-tech companies and those that depend on research and development. For example, many companies are now fighting over patents and copyrights on various smartphone and mobile technologies.

Like buildings and equipment, an intangible asset is recorded at its acquisition cost. However, unlike PPE, *IAS 38—Intangible Assets* states that it is unlikely that any subsequent expenditure will be recognized in the carrying amount of an intangible asset. The choice of cost model or revaluation model is also available for intangible assets, but *IAS 38* made it clear that it will be very rare for an entity to be able to use the revaluation model. In our discussions, we will assume that the intangibles will be measured at cost subsequent to acquisition.

The accounting for intangible assets can be a little more abstract. After all, it's definitely harder to account for things that have no physical form than those with physical form! We will cover the basics here, but you will likely revisit this topic in greater detail in more advanced accounting courses. Let's see what *IAS 38* says. Intangible assets fall into two categories:

- Intangibles with *finite lives* that can be measured reliably—we record **amortization** for these intangibles. Amortization works like depreciation and is usually computed on a straight-line basis. The residual value of most intangibles is zero. Intangibles with finite lives are also subjected to impairment tests.
- Intangibles with *indefinite lives*—because they have indefinite lives, we will not be able to allocate any amortization, so these intangibles are not amortized. Instead, they are tested for impairment for any loss in value, and, if any, a loss is recorded when it occurs. Goodwill is the most prominent example of an intangible asset with an indefinite life.

As far as intangible assets are concerned, remember that the opposite of *finite lives* is *indefinite lives*, and not *infinite lives*! Both categories of intangibles are subject to impairment tests. Impairment was previously explained in relation to PPE, but the same principles apply. For more details, if required by your course coverage, your instructor may refer you to *IAS 36—Impairment of Assets*. In the following discussions, we illustrate the accounting for both categories of intangibles.

This is what Airbus says about its intangible assets:

ADAPTED EXCERPTS FROM AIRBUS GROUP'S NOTES TO THE ACCOUNTS

Intangible Assets

Intangible assets comprise (1) goodwill, (2) capitalized development costs, and (3) other intangible assets, e.g., internally developed software and acquired intangible assets. Intangible assets with finite useful lives are generally amortized on a straight-line basis over their respective estimated useful lives (3 to 10 years) to their estimated residual values.

Source: Airbus Group's 2015 Financial Statements, page 34.

In the following discussions, we will look at some specific examples of intangible assets.

Accounting for Specific Intangibles

Each type of intangible asset is unique, and the accounting can vary from one asset to another. How the asset is acquired, for example, through purchase, a business combination (i.e., merger or acquisition of another company), or internal development, may also impact how the intangible asset is recognized and measured.

Patents. Patents are granted by a government to give the holder the exclusive right for a certain number of years to produce and sell an invention. The invention may be a specific product or process—for example, Apple's iPad and the Dolby noise-reduction process. Like any other asset, a patent may be purchased. Suppose Yamaha pays 15 million JPY (Japanese yen), or about \$170,000, to acquire a patent on January 1, and the business believes the expected useful life of the patent is five years (not necessarily the entire legal enforceability of the patent). Amortization expense is \$34,000 per year (\$170,000/5 years). Yamaha records the acquisition and amortization for this patent as follows:

A1				
	A	B	C	D
1	Jan. 1	Patents	170,000	
2		Cash		170,000
3		<i>To acquire a patent.</i>		
4				

A1				
	A	B	C	D
1	Dec. 31	Amortization Expense—Patents (\$170,000/5)	34,000	
2		Accumulated Amortization—Patents		34,000
3		<i>To amortize the cost of a patent.</i>		
4				

Alternatively, we can credit the patents account directly (not using an Accumulated Amortization account). Either way, the impact on the accounting equation is the same.

Assets		=	Liabilities		+	Shareholders' Equity	
- 34,000		=	0		+	- 34,000	(amortization expense)

Amortization for an intangible decreases both assets and equity exactly as depreciation does for equipment or a building.



Copyrights. Copyrights are exclusive rights to reproduce and sell a book, musical composition, film, or other work of art. Copyrights also protect computer software programs, such as Microsoft's Windows® and Excel. Issued by governments, copyrights in certain jurisdictions can extend up to 70 years beyond the author's (composer's, artist's, or programmer's) life. The cost of obtaining a copyright from the government is low, but a company may pay a large sum to purchase an existing copyright from the owner or to buy a company for its copyrights. For example, Zynga (one of the biggest social network game companies) has been buying game developers such as Newtoy (publisher of "Words with Friends") and OMGPOP (publisher of "Draw Something").

Trademarks and Trade Names. Trademarks and trade names (or *brand names*) are distinctive identification of a product or service. The "swoosh" symbol is the trademark of Nike, a symbol that has become synonymous with the company itself. Similarly, other logos such as Coca-Cola's "Dynamic Ribbon" and McDonald's golden arches and its *I'm lovin' it!* slogan are all registered trademarks. Often, you see distinctive identifications of products or services, marked with the symbol TM or ®. Some trademarks may have a definite useful life set by contract. Again, useful life for the purpose of amortization may be shorter than the legal useful life. This is similar to how the useful lives of a PPE, as intended by management, may be shorter than its actual capability.

Franchises and Licenses. Franchises and licenses are privileges granted by a private business or a government to sell a product or service in accordance with specified conditions. Many franchises are in the food or retail industry (Subway, Pizza Hut, Dunkin' Donuts, 7-Eleven, and thousands more), but there are also franchises in the service industry (H&R Block, Kumon Maths & Reading Centers, Days Inn, etc.). The useful lives of many franchises and licenses are usually indefinite (as the management is able to continue renewing the agreement) and, therefore, are not amortized.

Goodwill. In accounting, goodwill has a very specific meaning. Goodwill is defined as the excess of the cost of purchasing another company over the sum of the fair values of the acquired company's net assets (assets minus liabilities). For example, a purchaser is willing to pay for goodwill when the purchaser buys another company that has abnormal earning power to bring about a comparative advantage in the market. IAS 38 prohibits the recognition of internally generated goodwill. Thus, the only goodwill you will see in financial statements is goodwill that results from business combinations or mergers and acquisitions as outlined in *IFRS 3—Business Combinations*.

Airbus operates in many countries across Asia. Suppose Airbus acquires Europa Company at a cost of \$10 million. Europa's assets have a market value of \$9 million, and its liabilities total \$2 million, so Europa's net assets total \$7 million at current market value. Note that the assets and liabilities may include new identifiable intangible assets that were previously not recorded in Europa's books, for example brand names, trademarks, customer relationships, etc. In this case, Europa paid \$3 million for goodwill, computed as follows:

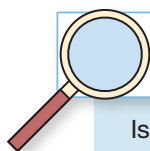
Purchase price paid for Europa Company		\$10 million
Sum of the fair values of Europa Company's assets.....	\$9 million	
Less: Europa Company's liabilities	(2 million)	
Market value of Europa Company's net assets		7 million
Excess is called <i>goodwill</i>		<u>\$ 3 million</u>

Airbus's entry to record the acquisition of Europa Company, including its goodwill, would be:

A1				
	A	B	C	D
1		Assets (Cash, Receivables, Inventories, PPE,		
2		all at fair value)	9,000,000	
3		Goodwill	3,000,000	
4		Liabilities		2,000,000
5		Cash		10,000,000
6				

Goodwill in accounting has special features, as follows:

1. Goodwill is recorded only when it is purchased in the acquisition of another company. A purchase transaction provides objective evidence of the value of goodwill as the excess of purchase consideration over the net fair value of identifiable assets and liabilities. Companies are not allowed to record internally generated goodwill that they create for their own business.
2. Unlike other intangibles with finite useful lives, goodwill is not amortized and is subjected to strict impairment tests.



A Closer Look

Is it possible to acquire a company for a consideration that is less than the fair value of its identifiable assets and liabilities? A “negative goodwill?” Yes! This is what *IFRS 3—Business Combinations* calls “bargain purchase.” Any resulting gain, subject to a number of caveats, may be recognized by the acquirer as a gain in the Income Statement on the acquisition date.

Accounting for Research and Development Costs

In a company with significant research and development (R&D) activities, such as Lenovo or Samsung, should the expenditures be treated as assets or expenses? Under *IAS 38*, the accounting treatment for R&D expenditures is literally split in the middle between research and development. Under *IFRS*, costs associated with the creation of intangible assets are classified into *research phase* costs and *development phase* costs. Costs in the research phase are always expensed. However, costs in the development phase are capitalized if the company can demonstrate meeting all of the following six criteria:

- the technical feasibility of completing the intangible asset
- the intention to complete the intangible asset
- the ability to use or sell the intangible asset
- the future economic benefits (e.g., the existence of a market or, if for internal use, the usefulness of the intangible asset)
- the availability of adequate resources to complete development of the asset
- the ability to reliably measure the expenditure attributable to the intangible asset during its development

Like many accounting standards, the separation between research and development is one that requires judgment, supported by objective evidence.

You saw earlier at the start of this chapter that one of Airbus’s intangible assets was “Capitalized Development Costs” (line 22), with a gross amount of €2,686 million, an accumulated amortization and impairment of €1,027 million, and, thus, a net carrying amount of €1,659 million. Its accounting policies reflect the discussion we had earlier about accounting for research and development costs.

ADAPTED EXCERPTS FROM AIRBUS GROUP’S NOTES TO THE ACCOUNTS

Research and Development

The costs for self-initiated research are expensed when incurred. The costs for self-initiated development are capitalized when:

- the product or process is technically feasible and clearly defined (i.e., the critical design review is finalized)

- adequate resources are available to successfully complete the development
- the benefits from the assets are demonstrated (a market exists or the internal usefulness is demonstrated), and the costs attributable to the projects are reliably measured
- the Group intends to produce and market or use the developed product or process and can demonstrate its profitability

Development costs which are capitalized are recognized either as intangible assets or, when the related development activities lead to the construction, of specialized tooling for production (“jigs and tools”), or involve the design, construction, and testing of prototypes and models, as property, plant, and equipment. Capitalized development costs are generally amortized over the estimated number of units produced. If the number of units produced cannot be estimated reliably, capitalized development costs are amortized over the estimated useful life of the internally generated intangible asset. Amortization of capitalized development costs is recognized in cost of sales.

Source: Airbus Group’s 2015 Financial Statements, page 15

Accounting for the Impairment of an Intangible Asset

Impairment testing applies to intangible assets as it does to PPE (as described earlier). Some intangibles—such as goodwill, licenses, and some trademarks—have indefinite lives and therefore are not subject to amortization. But all intangibles are subject to a write-down when their value decreases. Recall that such a decline in value of an asset is called impairment.

Let’s look at our hypothetical purchase of Europa Company by Airbus. After a couple of years, market conditions indicate that the goodwill paid during the acquisition of Europa Company is impaired. The recoverable amount of the goodwill is calculated to be \$1,000,000. Airbus would then record a \$2 million impairment loss and write down the book value of the goodwill, as follows:

A1				
	A	B	C	D
1		Impairment Loss on Goodwill (\$3 – \$1 million)	2 million	
2		Goodwill		2 million
3				

Both assets (goodwill) and equity decrease (through the Loss account). Unlike tangible assets, once goodwill is impaired, IFRS prohibit any reversal of the impairment. Airbus’s financial statements will report the following (in millions):

Assets	=	Liabilities	+	Shareholders’ Equity	
– 2 million	=	0	+	– 2 million	(impairment loss)

CASH FLOW



Reporting Long-Term Assets Transactions on the Statement of Cash Flows

Three main types of long-term asset transactions appear on the statement of cash flows: their acquisitions and disposals that are settled in cash. Specifically, acquisitions and disposal of long-term assets are cash flows from investing activities. Have a look Exhibit 7-13, which shows extracts of Airbus’s statements of cash flows for 2015 and 2014. Acquisitions and disposals of PPE, intangibles, assets, and investment property are denoted in color (lines 6 to 7).

Exhibit 7-13 Airbus's Statement of Cash Flows

A1			
	A	B	C
1	Airbus Group		
2	Consolidated Statement of Cash Flows (Adapted)		
3	For year ended December 31		
4	(In millions of Euro)	2015	2014
5	Cash flows from operating activities	3,600	2,560
6	Cash flows from investing activities		
7	Purchase of intangible assets, PPE and investment property	(2,924)	(2,548)
8	Process from disposal of intangible assets, PPE and investment property	78	232
9	Other cash flows from investing activities	(613)	(907)
10	Cash flows from investing activities	(3,459)	(3,223)
11	Cash flows from financing activities	(25)	495
12	Net cash flows (excluding foreign exchange rate changes)	116	(168)

As you can see, the cash components of long-term asset transactions will be reflected in the statement of cash flows. We will look into cash flows later in Chapter 11, but for now, just remember that the cash elements of PPE and intangibles transactions will be reported in the cash flow statement.

EVALUATE A COMPANY'S PERFORMANCE BASED ON ITS ASSETS

Owners, managers, and investors use ratios to evaluate a business. Two ratios relate directly to non-current assets: asset turnover ratio and return on asset. The ability of a business to generate sales and profit from its asset is an important indication of effectiveness. Non-productive assets will hurt asset turnover and return on assets.

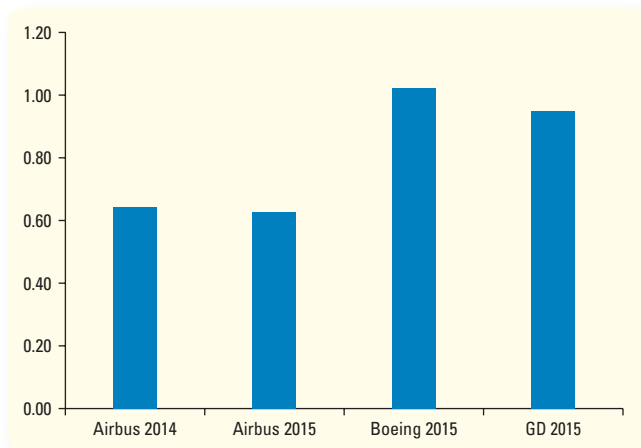
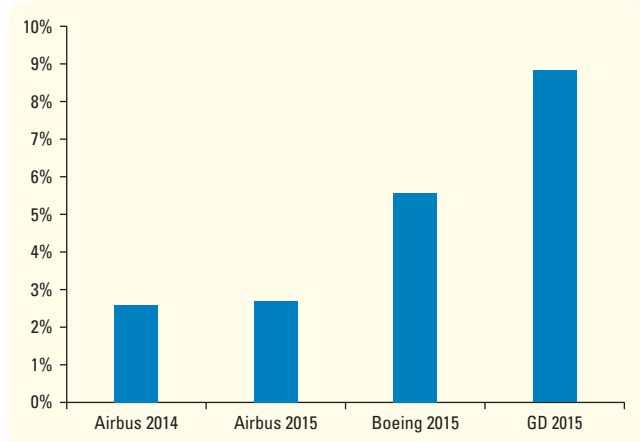
7 Evaluate a company's performance based on its assets

Asset Turnover and Return on Assets. **Asset turnover** is how many sales a business can generate from its assets. Similarly, **return on assets** is how much profit a business can generate from its assets. The higher the turnover or the return, the more effective the business is at utilizing its assets. Let's have a look at how Airbus performs on these two ratios and compare it to two other aircraft manufacturers, Boeing and General Dynamics (makers of the Gulfstream aircrafts, but also a defense contractor).

Asset Turnover and Return on Assets	Airbus 2015	Airbus 2014
$\frac{\text{Sales}}{\text{Average assets}^*}$	$\frac{64,450}{101,392} = 0.64 \text{ times}$	$\frac{60,713}{93,188} = 0.65$
$\frac{\text{Net profit}}{\text{Average assets}^*}$	$\frac{2,698}{101,392} = 2.66\%$	$\frac{2,350}{93,188} = 2.52$

*Average = (beginning + ending balance)/2

Airbus's asset turnover declined slightly (from 0.65 to 0.64 times), but its return on assets showed marginal improvements (from 2.52% to 2.66%) in 2015 compared to 2014. Exhibits 7-14 and 7-15 plot the two ratios for the three aircraft manufacturers. You can see that Airbus is behind Boeing and General Dynamics in terms of utilization of assets to generate revenue, and its returns are half that of Boeing's and one-third that of General Dynamics.

Exhibit 7-14 Asset Turnover Comparisons**Exhibit 7-15** Return on Assets Comparisons

DECISION GUIDELINES

PPE AND RELATED EXPENSES

Airbus, like all other companies, must make some decisions about how to account for its property, plant, and equipment (PPE) and intangibles. Let's review some of these decisions.

Decision

Capitalize or expense a cost?

Capitalize or expense:

- Cost associated with a new asset?
- Cost associated with an existing asset?

Which depreciation method to use:

- For financial reporting?
- For income tax?
- How to account for natural resources?
- How to account for intangibles?

Guidelines

General rule: Capitalize all costs that provide future benefit for the business such as a new package-handling system. Expense all costs that provide no future benefit, such as ordinary repairs to a delivery vehicle.

Capitalize all costs that bring the asset to its intended use, including asset purchase price, transportation charges, and taxes paid to acquire the asset.

Capitalize only those costs that add to the asset's usefulness or to its useful life. Expense all other costs as maintenance or repairs.

Use the method that best matches depreciation expense against the revenues produced by the asset. Most companies use the straight-line method.

Use the method that produces the fastest tax deductions. Depending on the applicable tax regulations, a company may be able to apply different depreciation methods for financial reporting and for income-tax purposes.

Capitalize the asset's acquisition cost and all later costs that add to the natural resource's future benefit. Then record depletion expense, as computed by the units-of-production method.

Capitalize acquisition cost for intangibles that meet the definition, recognition, and measurement criteria. For intangibles with finite lives, record amortization expense. For intangibles with indefinite lives, do not record amortization. All intangible assets are subject to impairment tests.

End-of-Chapter

Summary Problem

The figures that follow appear in the *Answers to the Mid-Chapter Summary Problem*, Requirement 2.

Financial Year	Method A: Straight-Line			Method B: Double-Declining-Balance		
	Annual Depreciation Expense	Accumulated Depreciation	Book Value	Annual Depreciation Expense	Accumulated Depreciation	Book Value
Start			\$44,000			\$44,000
20X6	\$4,000	\$ 4,000	40,000	\$8,800	\$ 8,800	35,200
20X7	4,000	8,000	36,000	7,040	15,840	28,160
20X8	4,000	12,000	32,000	5,632	21,472	22,528

Requirements

1. Suppose the income tax authorities permitted a choice between these two depreciation methods. Which method would Airbus select for income-tax purposes? Why?
2. Suppose Airbus purchased the equipment described earlier (€44,000 cost, €4,000 residual value, and 10 years' useful life) on January 1, 20X6. Management has depreciated the equipment by using the double-declining-balance method. On June 30, 20X8, Airbus sold the equipment for €27,000 cash.

Record depreciation for 20X7 and the sale of the equipment on June 30, 20X8.

Answers

Requirement 1

For tax purposes, most companies select the accelerated method because it results in the greatest depreciation in the earliest years of the asset's life. Accelerated depreciation minimizes income tax payments in the early years of the asset's life. That maximizes the business's cash at the earliest possible time.

Requirement 2

Entries to record depreciation to date of sale, and then the sale of the equipment, follow:

A1			
	A	B	D
1	20X8		
2	Jun 30	Depreciation Expense—Equipment ($\$5,632 \times 1/2$ year)	2,816
3		Accumulated Depreciation—Equipment	2,816
4		To update depreciation.	
5			
6		Cash	27,000
7	Jun 30	Accumulated Depreciation—Equipment ($\$15,840 + \$2,816$)	18,656
8		Equipment	44,000
9		Gain on Sale of Equipment	1,656
10		To record sale of equipment.	
11			

REVIEW | PPES and Intangibles

Quick Check (Answers are given at the end of the chapter.)

1. Bretman, Inc., purchased a tract of land, a small office building, and some equipment for \$1,800,000. The appraised value of the land was \$1,420,000, the building \$650,000, and the equipment \$430,000. What is the cost of the land?
 - a. \$600,000
 - b. \$1,022,400
 - c. \$1,420,000
 - d. None of the above
2. Which statement is false?
 - a. Depreciation is based on the matching principle because it matches the cost of the asset with the revenue generated over the asset's useful life.
 - b. Depreciation is a process of allocating the cost of a PPE over its useful life.
 - c. Depreciation creates a fund to replace the asset at the end of its useful life.
 - d. The cost of a PPE minus accumulated depreciation equals the asset's book value.

Use the following data for questions 3–6.

On July 1, 20X6, Amir Communications purchased a new piece of equipment that cost \$65,000. The estimated useful life is 10 years and estimated residual value is \$5,000.

3. What is the depreciation expense for 20X6 if Amir uses the straight-line method?
 - a. \$3,000
 - b. \$6,000
 - c. \$3,250
 - d. \$6,500
4. Assume Amir Communications purchased the equipment on January 1, 20X6. If Amir uses the straight-line method for depreciation, what is the asset's book value at the end of 20X7?
 - a. \$54,000
 - b. \$59,000
 - c. \$48,000
 - d. \$53,000
5. Assume Amir Communications purchased the equipment on January 1, 20X6. If Amir uses the double-declining-balance method, what is the depreciation for 20X7?
 - a. \$13,000
 - b. \$10,400
 - c. \$12,000
 - d. \$9,600
6. Return to Amir's original purchase date of July 1, 20X5. Assume that Amir uses the straight-line method of depreciation and sells the equipment for \$44,500 on July 1, 20X9. The result of the sale of the equipment is a gain (loss) of
 - a. \$3,500.
 - b. \$8,500.
 - c. \$ (15,500).
 - d. \$0.
7. A company bought a new machine for \$30,000 on January 1. The machine is expected to last five years and has a residual value of \$6,000. If the company uses the double-declining-balance method, accumulated depreciation at the end of year 2 will be
 - a. \$12,000.
 - b. \$9,600.
 - c. \$19,200.
 - d. \$15,360.
8. Which of the following is *not* a capital expenditure?
 - a. A complete overhaul of an air-conditioning system
 - b. Replacement of an old motor with a new one in a piece of equipment
 - c. The cost of installing a piece of equipment
 - d. The addition of a building wing
 - e. A tune-up of a company vehicle

9. Which of the following assets is *not* subject to a decreasing book value through depreciation, depletion, or amortization?
 - a. Goodwill
 - b. Land improvements
 - c. Natural resources
 - d. Intangibles
10. Why would a business select an accelerated method of depreciation for tax purposes?
 - a. Accelerated depreciation will result in higher gain on disposal of PPE than straight-line depreciation.
 - b. Accelerated depreciation generates a greater amount of depreciation over the life of the asset than does straight-line depreciation.
 - c. Accelerated depreciation is easier to calculate because salvage value is ignored.
 - d. Accelerated depreciation generates higher depreciation expense immediately, and therefore lowers tax payments in the early years of the asset's life.
11. A company purchased an oil well for \$270,000. It estimates that the well contains 100,000 barrels, has an eight-year life, and has no salvage value. If the company extracts and sells 20,000 barrels of oil in the first year, how much depletion expense should be recorded?
 - a. \$33,750
 - b. \$54,000
 - c. \$27,000
 - d. \$67,500
12. Which item among the following is not an intangible asset?
 - a. A copyright
 - b. A patent
 - c. A trademark
 - d. Goodwill
 - e. All of the above are intangible assets.

Accounting Vocabulary

accelerated depreciation method (p. 404) A depreciation method that writes off a relatively larger amount of the asset's cost nearer the start of its useful life than does the straight-line method.

accumulated depreciation (p. 400) The cumulative sum of all depreciation expense from the date of acquiring PPE.

amortization (p. 418) The systematic reduction of a lump-sum amount. Expense that applies to intangible assets in the same way depreciation applies to PPE and depletion applies to natural resources.

asset turnover (p. 423) A measure of efficiency on the use of assets to generate sales.

capital expenditure (p. 397) An expenditure that increases an asset's capacity or efficiency or extends its useful life. Capital expenditures are debited to an asset account.

copyright (p. 420) Exclusive right to reproduce and sell a book, musical composition, film, other work of art, or computer program. Issued by the government, copyrights may extend many years beyond the author's life.

depletion expense (p. 417) That portion of a natural resource's cost that is used up in a particular period. Depletion expense is computed in the same way as units-of-production depreciation.

depreciation (p. 399) The systematic allocation of the cost of a long-term asset over its useful life.

depreciable cost (p. 401) The cost of a PPE minus its estimated residual value.

double-declining-balance (DDB) method (p. 404) An accelerated depreciation method that computes annual depreciation by multiplying the asset's decreasing book value by a constant percentage, which is two times the straight-line rate.

estimated residual value (p. 401) Expected cash value of an asset at the end of its useful life. Also called *residual value*, *scrap value*, or *salvage value*.

estimated useful life (p. 401) Length of service that a business expects to get from an asset. May be expressed in years, units of output, miles, or other measures.

franchises and licenses (p. 420) Privileges granted by a private business or a government to sell a product or service in accordance with specified conditions.

goodwill (p. 420) Excess of the cost of an acquired company over the sum of the market values of its net assets (assets minus liabilities).

impairment (p. 391) The condition that exists when the carrying amount of an asset exceeds its recoverable amount.

intangible assets (p. 392) A non-monetary asset with no physical form that conveys future economic benefits to the entity.

patent (p. 419) A government grant giving the holder the exclusive right for a certain number of years to produce and sell an invention.

property, plant and equipment (PPE) (p. 391) Long-lived assets, such as land, buildings, and equipment, used in the operation of the business. Also called *fixed assets* or *plant assets*.

recoverable amount (p. 412) The higher of fair value less cost to sell and value-in-use. Used in impairment tests.

return on assets (p. 423) A measure of the level of returns or profit generated from assets.

straight-line (SL) method (p. 402) Depreciation method in which an equal amount of depreciation expense is assigned to each year of asset use.

trademark, trade name (p. 420) A distinctive identification of a product or service. Also called a *brand name*.

units-of-production (UOP) method (p. 403) Depreciation method by which a fixed amount of depreciation is assigned to each unit of output produced by the PPE.

ASSESS YOUR PROGRESS

Short Exercises

L0 2

S7-1. (Learning Objective 2: Determining cost and book value of a company's PPEs) Examine Round Rock's assets.

	A1		
		A	B
			C
1		Hard Stone Corporation	
2		Consolidated Balance Sheets (Partial, Adapted)	
3		(In millions)	May 31, 20X7
4		Assets	May 31, 20X6
5		Current assets	
6		Cash and cash equivalents	\$ 2,098 \$ 246
7		Receivables, less allowances of \$144 and \$125	2,772 2,610
8		Spare parts, supplies and fuel	4,670 4,510
9		Prepaid expenses and other	468 411
10		Total current assets	10,008 7,777
11		Property and equipment, at cost	
12		Aircraft	2,394 2,394
13		Package handling and ground support equipment	12,225 12,139
14		Computer and electronic equipment	28,165 26,115
15		Vehicles	586 453
16		Facilities and other	1,435 1,594
17		Total cost	44,805 42,695
18		Less: Accumulated depreciation	(14,903) (12,942)
19		Net property and equipment	29,902 29,753
20		Other long-term assets	
21		Goodwill	724 724
22		Prepaid pension cost	1,341 1,275
23		Intangible and other assets	324 329
24		Total other long-term assets	2,389 2,328
25		Total assets	\$ 42,299 \$ 39,858

1. What is Hard Stone's largest category of assets? List all 20X7 assets in the largest category and their amounts as reported by Hard Stone.
2. What was Hard Stone's cost of property and equipment at May 31, 20X7? What was the book value of property and equipment on this date? Why is book value less than cost?

L0 2

S7-2. (Learning Objective 2: Measuring the cost of a PPE) The costs included for the acquisition of land are listed in the "Land and Land Improvements" section of this chapter. First is the purchase price of the land, which is obviously included in the cost of the land. The reasons for including the other costs are not so obvious. For example, property tax is ordinarily an expense, not part of the cost of an asset. State why the other costs listed are included as part of the cost of the land. After the land is ready for use, will these related costs be capitalized or expensed?

S7-3. (Learning Objective 2: Determining the cost of individual assets in a lump-sum purchase of assets) Farah Distribution Service pays \$110,000 for a group purchase of land, building, and equipment. At the time of acquisition, the land has a current market value of \$75,000, the building's current market value is \$45,000, and the equipment's current market value is \$30,000. Journalize the lump-sum purchase of the three assets for a total cost of \$110,000. You sign a note payable for this amount.

L0 2

S7-4. (Learning Objective 2: Capitalizing versus expensing PPE costs) Assume Speed Car Rentals changed the tires for its entire fleet of cars at a cost of \$1.8 million, which Speed paid in cash. Further, assume the Speed accountant erroneously capitalized this expense as part of the cost of the fleet.

L0 2

Show the effects of the accounting error on Speed Car Rentals' Income Statement. To answer this question, determine whether revenues, total expenses, and net income were overstated or understated by the accounting error.

S7-5. (Learning Objective 4: Computing depreciation by three methods—first year only) Assume that at the beginning of 20X6, AirAsia, a regional airline operating predominantly in Southeast Asia, purchased a used Boeing 737 aircraft at a cost of \$55,000,000. AirAsia expects the plane to remain useful for five years (7 million miles) and to have a residual value of \$6,000,000. AirAsia expects to fly the plane 875,000 miles the first year; 1,475,000 miles each year during the second, third, and fourth years; and 1,700,000 miles the last year.

L0 4

1. Compute AirAsia's first-year depreciation on the plane using the following methods:
 - a. Straight-line
 - b. Units-of-production
 - c. Double-declining-balance
2. Show the airplane's book value at the end of the first year under each depreciation method.

S7-6. (Learning Objective 4: Computing depreciation by three methods—third year only) Use the AirAsia data in Short Exercise 7-5 to compute AirAsia's third-year depreciation on the plane using the following methods:

L0 4

- a. Straight-line
- b. Units-of-production
- c. Double-declining-balance

S7-7. (Learning Objective 4: Selecting the best depreciation method for income tax purposes) This exercise uses the assumed AirAsia data from Short Exercise 7-5. Assume AirAsia is trying to decide which depreciation method to use for income tax purposes. The company can choose from among the following methods: (a) straight-line, (b) units of production, or (c) double-declining-balance.

L0 4

1. Which depreciation method offers the tax advantage for the first year? Describe the nature of the tax advantage.
2. How much income tax will AirAsia save for the first year of the airplane's use under the method you selected above as compared with using the straight-line depreciation method? Assume the tax rate is 30%. Ignore any earnings from investing the extra cash.

S7-8. (Learning Objectives 3, 4: Computing partial year depreciation; selecting the best depreciation method) Assume that on September 30, 20X6, LoganAir, the national airline of Switzerland, purchased an Airbus aircraft at a cost of €48,000,000. LoganAir expects the plane to remain useful for six years (4,500,000 miles) and to have a residual value of €6,000,000. LoganAir will fly the plane 420,000 miles during the remainder of 20X6. Compute LoganAir's depreciation on the plane for the year ended December 31, 20X6, using the following methods:

L0 3 4

- a. Straight-line
- b. Units-of-production
- c. Double-declining-balance

Which method would produce the highest net income for 20X6? Which method produces the lowest net income?

L0 3 4

S7-9. (Learning Objectives 3, 4: Computing and recording depreciation after a change in useful life of the asset) Ten Flags over Georgia paid \$120,000 for a concession stand. Ten Flags started out depreciating the building using the straight-line method over 20 years with zero residual value. After using the concession stand for three years, Ten Flags determines that the building will remain useful for only six more years. Record Ten Flags' depreciation on the concession stand for year 4 by the straight-line method.

L0 3 4

S7-10. (Learning Objectives 3, 4: Computing depreciation; recording a gain or loss on disposal) On January 1, 20X6, Scoot Airline purchased an airplane for \$38,700,000. Scoot Airline expects the plane to remain useful for six years and to have a residual value of \$3,500,000. Scoot Airline uses the straight-line method to depreciate its airplanes. Scoot Airline flew the plane for three years and sold it on January 1, 20X9, for \$9,300,000.

1. Compute accumulated depreciation on the airplane at January 1, 20X9 (same as December 31, 20X8).
2. Record the sale of the plane on January 1, 20X9.

L0 5

S7-11. (Learning Objective 5: Accounting for the depletion of a company's natural resources) Abundant Petroleum, the giant oil company, holds reserves of oil and gas assets. At the end of 20X6, assume the cost of Abundant Petroleum's mineral assets totaled \$160 billion, representing 10 billion barrels of oil in the ground.

1. Which depreciation method is similar to the depletion method that Abundant Petroleum and other oil companies use to compute their annual depletion expense for the minerals removed from the ground?
2. Suppose Abundant Petroleum removed 0.4 billion barrels of oil during 20X7. Record depletion expense for the year. Show amounts in billions.
3. At December 31, 20X6, Abundant Petroleum's Accumulated Depletion account stood at \$58 billion. Report Mineral Assets and Accumulated Depletion at December 31, 20X7. Do Abundant Petroleum's Mineral Assets appear to be plentiful or mostly used up? Give your reason.

L0 1 6

S7-12. (Learning Objectives 1, 6: Measuring and recording goodwill; Reporting cash flows) Hector, Inc., dominates the snack-food industry with its Tasty-Chip brand. Assume that Hector, Inc., purchased Crunchy Snacks, Inc., for \$10.8 million cash. The market value of Concord Snacks' assets is \$18 million, and Concord Snacks has liabilities of \$12 million.

1. Compute the cost of the goodwill purchased by Hector.
2. Explain how Hector will account for goodwill in future years.

L0 5

S7-13. (Learning Objective 5: Accounting for patents and research and development costs) This exercise summarizes the accounting for patents, which, like copyrights, trademarks, and franchises, provide the owner with a special right or privilege. It also covers research and development costs.

Suppose Solar Automobiles Limited paid \$700,000 to research and develop a new global positioning system. Solar also paid \$420,000 to acquire a patent on a new motor. After readying the motor for production, Solar's sales revenue for the first year totaled \$5,600,000. Cost of goods sold was \$3,900,000, and selling expenses totaled \$580,000. All these transactions occurred during 20X6. Solar expects the patent to have a useful life of seven years.

Prepare Solar Automobiles' Income Statement for the year ended December 31, 20X6, complete with a heading. Ignore income tax.

L0 7

S7-14. (Learning Objective 7: Reporting investing activities on the statement of cash flows) During 20X6, Orion Satellite Systems, Inc., purchased two other companies for \$18 million. Also during 20X6, Orion made capital expenditures of \$8 million to expand its market share. During the year, Orion sold its North American operations, receiving cash of \$16 million. Overall, Orion reported a net income of \$3 million during 20X6.

Show what Orion would report for cash flows from investing activities on its statement of cash flows for 20X6. Report a total amount for net cash provided by (used in) investing activities.

Exercises MyLab Accounting

Select A and B exercises can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select exercises using MyLab Accounting.

Group A

E7-15A. (Learning Objective 1: Determining the cost of PPE) Ariana Self Storage purchased land, paying \$185,000 cash as a downpayment and signing a \$200,000 note payable for the balance. Ariana also had to pay delinquent property tax of \$4,500, title insurance costing \$3,000, and \$10,000 to level the land and remove an unwanted building. The company paid \$63,000 to add soil for the foundation and then constructed an office building at a cost of \$640,000. It also paid \$57,000 for a fence around the property, \$15,000 for the company sign near the property entrance, and \$9,000 for lighting of the grounds. Determine the cost of Ariana's land, land improvements, and building.

L0 1

E7-16A. (Learning Objectives 1, 2: Allocating costs to assets acquired in a lump-sum purchase; disposing of a PPE) Goodwood Manufacturing bought three used machines in a \$154,000 lump-sum purchase. An independent appraiser valued the machines as shown in the table.

L0 1 2

Machine No.	Appraised Value
1	\$38,250
2	73,100
3	58,650

What is each machine's individual cost? Immediately after making this purchase, Goodwood sold machine 2 for its appraised value. What is the result of the sale? (Round decimals to three places when calculating proportions, and use your computed percentages throughout.)

E7-17A. (Learning Objective 1: Distinguishing capital expenditures from expenses) Assume Sweets Emporium, Inc., purchased conveyor-belt machinery. Classify each of the following expenditures as a capital expenditure or an immediate expense related to machinery:

L0 1

- Income tax paid on income earned from the sale of products manufactured by the machinery
- Major overhaul to extend the machinery's useful life by three years
- Ordinary repairs to keep the machinery in good working order
- Lubrication of the machinery before it is placed in service
- Periodic lubrication after the machinery is placed in service
- Sales tax paid on the purchase price
- Transportation and insurance while machinery is in transit from seller to buyer
- Purchase price
- Installation
- Training of personnel for initial operation of the machinery
- Special reinforcement to the machinery platform

E7-18A. (Learning Objective 4: Measuring, depreciating, and reporting PPE) During 20X6, Cho Book Store paid \$485,000 for land and built a store in Akron. Prior to construction, the city of Akron charged Cho \$1,500 for a building permit, which Cho paid. Cho also paid \$15,200 for architect's fees. The construction cost of \$670,000 was financed by a long-term note payable, with interest cost of \$27,300 paid at completion of the project. The building was completed September 30, 20X6. Cho depreciates the building by the straight-line method over 35 years, with estimated residual value of \$338,000.

L0 4

1. Journalize transactions for:
 - a. Purchase of the land
 - b. All the costs chargeable to the building in a single entry
 - c. Depreciation on the building
 Explanations are not required.
2. Report Cho Book Store's PPE on the company's Balance Sheet at December 31, 20X6.
3. What will Cho's Income Statement for the year ended December 31, 20X6, report for this situation?

LO 3 4

■ writing assignment

■ spreadsheet

E7-19A. (Learning Objectives 3, 4: Determining depreciation amounts by three methods; depreciation) West Side Pizza bought a used Nissan delivery van on January 2, 20X6, for \$19,200. The van was expected to remain in service for four years (32,000 miles). At the end of its useful life, West Side's officials estimated that the van's residual value would be \$3,200. The van traveled 10,000 miles the first year, 12,000 miles the second year, 4,000 miles the third year, and 6,000 miles the fourth year. Prepare a schedule of *depreciation expense* per year for the van under the three depreciation methods. (For units-of-production and double-declining-balance, round to the nearest two decimals after each step of the calculation.)

Which method best tracks the wear and tear on the van? Which method would West Side prefer to use for income tax purposes? Explain in detail why West Side prefers this method.

LO 1 4 7

E7-20A. (Learning Objectives 1, 4, 7: Reporting PPE; depreciation; and investing cash flows) Assume that in January 20X6, an Hotcake House restaurant purchased a building, paying \$57,000 cash and signing a \$108,000 note payable. The restaurant paid another \$60,000 to remodel the building. Furniture and fixtures cost \$54,000, and dishes and supplies—a current asset—were obtained for \$10,200.

Hotcake House is depreciating the building over 20 years by the straight-line method, with estimated residual value of \$55,000. The furniture and fixtures will be replaced at the end of five years and are being depreciated by the double-declining-balance method, with zero residual value. At the end of the first year, the restaurant still has dishes and supplies worth \$1,900.

Show what the restaurant will report for supplies, PPE, and cash flows at the end of the first year on its:

- Income Statement
- Balance Sheet
- Statement of cash flows (investing only)

Note: The purchase of dishes and supplies is an operating cash flow because supplies are a current asset.

LO 2

E7-21A. (Learning Objective 2: Selecting the best depreciation method for income tax purposes) On June 30, 20X6, Rafa Corp. paid \$260,000 for equipment that is expected to have an eight-year life. In this industry, the residual value of equipment is approximately 10% of the asset's cost. Rafa's cash revenues for the year are \$125,000, and cash expenses total \$75,000.

Assume Rafa has a choice of straight-line or DDB depreciation for taxation purposes. Select the depreciation method for income tax purposes. Then determine the extra amount of cash that Rafa can invest by using DDB depreciation, versus straight-line, for the year ended December 31, 20X6. The income tax rate is 40%.

LO 2

E7-22A. (Learning Objective 2: Changing a PPE's useful life) Assume G-1 Designing Consultants purchased a building for \$420,000 and depreciated it on a straight-line basis over 40 years. The estimated residual value was \$56,000. After using the building for 20 years, G-1 realized that the building will remain useful for only 15 more years. Starting with the 21st year, G-1 began depreciating the building over a revised total life of 35 years and decreased the residual value to \$13,000. Record depreciation expense on the building for years 20 and 21.

LO 2 3

E7-23A. (Learning Objectives 2, 3: Analyzing the effect of a sale of a PPE; DDB depreciation) Assume that on January 2, 20X6, Maxwell of Michigan purchased fixtures for \$8,900 cash, expecting the fixtures to remain in service for five years. Maxwell has depreciated the

fixtures on a double-declining-balance basis, with \$1,400 estimated residual value. On August 31, 20X7, Maxwell sold the fixtures for \$2,800 cash. Record both the depreciation expense on the fixtures for 20X7 and the sale of the fixtures. Apart from your journal entry, also show how to compute the gain or loss on Maxwell's disposal of these fixtures.

E7-24A. (Learning Objectives 1, 2, 3: Measuring a PPE's cost; using UOP depreciation; trading in a used asset) Prompt Truck Company is a large trucking company that operates throughout the United States. Prompt Truck Company uses the units-of-production (UOP) method to depreciate its trucks.

L0 1 2 3

Prompt Truck Company trades in trucks often to keep driver morale high and to maximize fuel economy. Consider these facts about one Mack truck in the company's fleet. When acquired in 20X6, the tractor-trailer rig cost \$390,000 and was expected to remain in service for 10 years or 1,000,000 miles. Estimated residual value was \$100,000. During 20X6, the truck was driven 78,000 miles; during 20X7, 118,000 miles; and during 20X8, 156,000 miles. After 38,000 miles in 20X9, the company traded in the Mack truck for a less expensive Freightliner with a sticker price of \$300,000. Prompt Truck Company paid cash of \$27,000. Determine Prompt's gain or loss on the transaction. Prepare the journal entry to record the trade-in of the old truck for the new one.

E7-25A. (Learning Objective 4: Recording natural resource assets and depletion) Rich Mines paid \$428,000 for the right to extract ore from a 275,000-ton mineral deposit. In addition to the purchase price, Rich Mines also paid a \$180 filing fee, a \$2,120 license fee to the state of Colorado, and \$65,000 for a geologic survey of the property. Because the company purchased the rights to the minerals only, it expects the asset to have zero residual value when fully depleted. During the first year of production, Rich Mines removed 45,000 tons of ore. Make journal entries to record (a) purchase of the mineral rights, (b) payment of fees and other costs, and (c) depletion for first-year production. What is the mineral asset's book value at the end of the year?

L0 4

E7-26A. (Learning Objectives 3, 5: Recording intangibles, amortization, and a change in the asset's useful life)

L0 3 5

1. Master Printers purchased for \$920,000 a patent for a new laser printer. Although the patent gives legal protection for 20 years, it is expected to provide Master Printers with a competitive advantage for only 10 years. Assuming the straight-line method of amortization, make journal entries to record (a) the purchase of the patent and (b) amortization for year 1.
2. After using the patent for five years, Master Printers learns at an industry trade show that Super Printers is designing a more efficient printer. On the basis of this new information, Master Printers determines that the patent's total useful life is only seven years. Record amortization for year 6.

E7-27A. (Learning Objective 5: Computing and accounting for goodwill) Assume Holden paid \$15 million to purchase Northshore.com. Assume further that Northshore had the following summarized data at the time of the Holden acquisition (amounts in millions):

L0 5

Northshore.com			
Assets		Liabilities and Equity	
Current assets	\$13	Total liabilities	\$25
Long-term assets	23	Shareholders' equity	11
	<u>\$36</u>		<u>\$36</u>

Northshore's long-term assets had a current market value of only \$20 million.

Requirements

1. Compute the cost of goodwill purchased by Holden.
2. Journalize Holden's purchase of Northshore.
3. Explain how Holden will account for goodwill in the future.

L0 6

E7-28A. (Learning Objective 6: Reporting cash flows for property and equipment) Assume Shoe Warehouse Corporation completed the following transactions:

- Sold a store building for \$680,000. The building had cost Shoe Warehouse \$1,800,000, and at the time of the sale its accumulated depreciation totaled \$1,100,000.
- Lost a store building in a fire. The building cost \$390,000 and had accumulated depreciation of \$180,000. The insurance proceeds received by Shoe Warehouse totaled \$140,000.
- Renovated a store at a cost of \$170,000.
- Purchased store fixtures for \$80,000. The fixtures are expected to remain in service for 10 years and then be sold for \$10,000. Shoe Warehouse uses the straight-line depreciation method.

For each transaction, show what Shoe Warehouse would report for investing activities on its statement of cash flows. Show negative amounts in parentheses.

Group B

L0 1

E7-29B. (Learning Objective 1: Determining the cost of PPE) Deville Self Storage purchased land, paying €165,000 cash as a downpayment and signing a €192,000 note payable for the balance. Deville also had to pay delinquent property tax of €4,500, title insurance costing €3,000, and €6,000 to level the land and remove an unwanted building. The company paid €58,000 to add soil for the foundation and then constructed an office building at a cost of €620,000. It also paid €48,000 for a fence around the property, €22,000 for the company sign near the property entrance, and €2,000 for lighting of the grounds. Determine the cost of Deville's land, land improvements, and building.

L0 1 2

E7-30B. (Learning Objectives 1, 2: Allocating costs to assets acquired in a lump-sum purchase; disposing of a PPE) Clintwood Manufacturing bought three used machines in a €226,000 lump-sum purchase. An independent appraiser valued the machines as shown in the table.

Machine No.	Appraised Value
1	€ 84,000
2	127,200
3	28,800

What is each machine's individual cost? Immediately after making this purchase, Clintwood sold machine 2 for its appraised value. What is the result of the sale? (Round decimals to three places when calculating proportions, and use your computed percentages throughout.)

L0 1

E7-31B. (Learning Objective 1: Distinguishing capital expenditures from expenses) Assume Desserts Galore, Inc., purchased conveyor-belt machinery. Classify each of the following expenditures as a capital expenditure or an immediate expense related to machinery:

- Training of personnel for initial operation of the machinery
- Special reinforcement to the machinery platform
- Income tax paid on income earned from the sale of products manufactured by the machinery
- Major overhaul to extend the machinery's useful life by three years
- Sales tax paid on the purchase price
- Transportation and insurance while machinery is in transit from seller to buyer
- Purchase price
- Installation
- Lubrication of the machinery before it is placed in service
- Ordinary repairs to keep the machinery in good working order
- Periodic lubrication after the machinery is placed in service

E7-32B. (Learning Objectives 1, 2: Measuring, depreciating, and reporting PPE) During 20X6, Lao Book Store paid €489,000 for land and built a store in Lisbon. Prior to construction, the city of Lisbon charged Tao €1,600 for a building permit, which Lao paid. Lao also paid €15,700 for architect's fees. The construction cost of €210,000 was financed by a long-term note payable, with interest cost of €30,100 paid at completion of the project. The building was completed on September 30, 20X6. Lao depreciates the building by the straight-line method over 35 years, with estimated residual value of €344,000.

1. Journalize transactions for:
 - a. Purchase of the land
 - b. All the costs chargeable to the building in a single entry
 - c. Depreciation on the building
 Explanations are not required.
2. Report Lao Book Store's PPE on the company's Balance Sheet at December 31, 20X6.
3. What will Lao's Income Statement for the year ended December 31, 20X6, report for this situation?

E7-33B. (Learning Objectives 2, 3: Determining depreciation amounts by three methods) Southern's Pizza bought a used Nissan delivery van on January 2, 20X6, for €19,500. The van was expected to remain in service four years (30,000 miles). At the end of its useful life, Southern's officials estimated that the van's residual value would be €2,700. The van traveled 7,000 miles the first year, 9,500 miles the second year, 4,500 miles the third year, and 9,000 miles the fourth year. Prepare a schedule of *depreciation expense* per year for the van under the three depreciation methods. (For units-of-production and double-declining-balance, round to the nearest two decimals after each step of the calculation.)

Which method best tracks the wear and tear on the van? Which method would Southern's prefer to use for income tax purposes? Explain in detail why Southern's prefers this method.

E7-34B. (Learning Objectives 1, 2, 6: Reporting PPE; depreciation; and investing cash flows) Assume that in January 20X6, an Exquisite Eatery restaurant purchased a building, paying €53,000 cash and signing a €106,000 note payable. The restaurant paid another €65,000 to remodel the building. Furniture and fixtures cost €59,000, and dishes and supplies—a current asset—were obtained for €9,800.

Exquisite Eatery is depreciating the building over 20 years by the straight-line method, with estimated residual value of €52,000. The furniture and fixtures will be replaced at the end of five years and are being depreciated by the double-declining-balance method, with zero residual value. At the end of the first year, the restaurant still has dishes and supplies worth €1,500.

Show what the restaurant will report for supplies, PPE, and cash flows at the end of the first year on its:

- Income Statement
- Balance Sheet
- Statement of cash flows (investing only)

Note: The purchase of dishes and supplies is an operating cash flow because supplies are a current asset.

E7-35B. (Learning Objective 2: Selecting the best depreciation method for income tax purposes) On June 30, 20X6, Reel Corp. paid €240,000 for equipment that is expected to have an eight-year life. In this industry, the residual value is approximately 10% of the asset's cost. Reel's cash revenues for the year are €160,000 and cash expenses total €120,000.

Assume Reel has a choice of straight-line or DDB depreciation for taxation purposes. Select the depreciation method for income tax purposes. Then determine the extra amount of cash that Reel can invest by using DDB depreciation, versus straight-line, for the year ended December 31, 20X6. The income tax rate is 40%.

E7-36B. (Learning Objective 2: Changing a PPE's useful life) Assume B-1 Accounting Consultants purchased a building for €438,000 and depreciated it on a straight-line basis over 40 years. The estimated residual value was €72,000. After using the building for 20 years, B-1 realized that the building will remain useful for only 15 more years. Starting with the 21st year,

L0 1 2

L0 2 3

■ writing assignment

■ spreadsheet

L0 1 2 6

L0 2

L0 2

B-1 began depreciating the building over the newly revised total life of 35 years and decreased the estimated residual value to €13,800. Record depreciation expense on the building for years 20 and 21.

L0 2 3

E7-37B. (Learning Objectives 2, 3: Analyzing the effect of a sale of a PPE; DDB depreciation) Assume that on January 2, 20X6, LaSalle of Lyon purchased fixtures for €8,400 cash, expecting the fixtures to remain in service for five years. LaSalle has depreciated the fixtures on a double-declining-balance basis, with €1,800 estimated residual value. On September 30, 20X7, LaSalle sold the fixtures for €2,200 cash. Record both the depreciation expense on the fixtures for 20X7 and then the sale of the fixtures. Apart from your journal entry, also show how to compute the gain or loss on LaSalle's disposal of these fixtures.

L0 1 2 3

E7-38B. (Learning Objectives 1, 2, 3: Measuring a PPE's cost; using UOP depreciation; trading in a used asset) EasyTruck Company is a large trucking company that operates throughout the EU. EasyTruck Company uses the units-of-production (UOP) method to depreciate its trucks.

EasyTruck Company trades in trucks often to keep driver morale high and to maximize fuel economy. Consider these facts about one Mack truck in the company's fleet. When acquired in 20X6, the rig cost €360,000 and was expected to remain in service for 10 years or 1,000,000 miles. Estimated residual value was €100,000. During 20X6, the truck was driven 79,000 miles; during 20X7, 119,000 miles; and during 20X8, 158,000 miles. After 44,000 miles in 20X9, the company traded in the Mack truck for a less expensive Freightliner with a sticker price of €300,000. EasyTruck Company paid cash of €24,000. Determine EasyTruck's gain or loss on the transaction. Prepare the journal entry to record the trade-in of the old truck on the new one.

L0 4

E7-39B. (Learning Objective 4: Recording natural resource assets and depletion) Gold Mines paid €435,000 for the right to extract ore from a 400,000-ton mineral deposit. In addition to the purchase price, Gold Mines also paid a €170 filing fee, a €2,600 license fee, and €92,230 for a geologic survey of the property. Because the company purchased the rights to the minerals only, it expected the asset to have zero residual value when fully depleted. During the first year of production, Gold Mines removed 80,000 tons of ore. Make journal entries to record (a) the purchase of the mineral rights, (b) the payment of fees and other costs, and (c) the depletion for first-year production. What is the mineral asset's book value at the end of the year?

L0 3 5

E7-40B. (Learning Objectives 3, 5: Recording intangibles; amortization, and a change in the asset's useful life)

1. Royal Printers purchased for €780,000 a patent for a new laser printer. Although the patent gives legal protection for 20 years, it is expected to provide Royal Printers with a competitive advantage for only eight years. Assuming the straight-line method of amortization, make journal entries to record (a) the purchase of the patent and (b) amortization for year 1.
2. After using the patent for four years, Royal Printers learns at an industry trade show that Speedy Printers is designing a more efficient printer. On the basis of this new information, Royal Printers determines that the patent's total useful life is only six years. Record amortization for year 5.

L0 5

E7-41B. (Learning Objective 5: Computing and accounting for goodwill) Assume Golden paid €20 million to purchase Southwest.com. Assume further that Southwest had the following summarized data at the time of the Golden acquisition (amounts in millions):

Southwest.com			
Assets		Liabilities and Equity	
Current assets	€10	Total liabilities	€25
Long-term assets	22	Shareholders' equity	7
	<u>€32</u>		<u>€32</u>

Southwest's long-term assets had a current market value of only €18 million.

Requirements

1. Compute the cost of goodwill purchased by Golden.
2. Journalize Golden's purchase of Southwest.
3. Explain how Golden will account for goodwill in the future.

E7-42B. (Learning Objective 6: Reporting cash flows for property and equipment) Assume Shoes-R-Us Corporation completed the following transactions:

L0 6

- a. Sold a store building for €640,000. The building had cost Shoes-R-Us €1,400,000, and at the time of the sale its accumulated depreciation totaled €780,000.
- b. Lost a store building in a fire. The building cost €360,000 and had accumulated depreciation of €160,000. The insurance proceeds received by Shoes-R-Us totaled €120,000.
- c. Renovated a store at a cost of €140,000.
- d. Purchased store fixtures for €100,000. The fixtures are expected to remain in service for 10 years and then be sold for €20,000. Shoes-R-Us uses the straight-line depreciation method.

For each transaction, show what Shoes-R-Us would report for investing activities on its statement of cash flows. Show negative amounts in parentheses.

Challenge Exercises

E7-43. (Learning Objective 2: Computing units-of-production depreciation) Buff Gym purchased exercise equipment at a cost of \$116,000. In addition, Buff paid \$4,000 for a special platform on which to stabilize the equipment for use. Freight costs of \$1,800 to ship the equipment were borne by the seller. Buff will depreciate the equipment by the units-of-production method, based on an expected useful life of 60,000 hours of exercise. The estimated residual value of the equipment is \$12,000. How many hours did Buff Gym use the machine if depreciation expense is \$4,680?

L0 2

E7-44. (Learning Objective 4: Determining the sale price of property and equipment) Willis Corporation reported the following for property and equipment (in millions, adapted):

L0 4

	Year End	
	20X7	20X6
Property and equipment	\$24,073	\$22,011
Accumulated depreciation	(13,306)	(12,087)

During 20X7, Willis paid \$2,580 million for new property and equipment. Depreciation for the year totaled \$1,595 million. During 20X7, Willis sold property and equipment for cash of \$45 million. How much was Willis's gain or loss on the sale of property and equipment during 20X7?

E7-45. (Learning Objectives 2, 3: Determining net income after a change in depreciation method) Norzani, Inc., has a popular line of sunglasses. Norzani reported net income of \$68 million for 20X6. Depreciation expense for the year totaled \$36 million. Norzani, Inc., depreciated PPE over eight years using the straight-line method and no residual value.

L0 2 3

Norzani Inc., paid \$288 million for PPE at the beginning of 20X6. Then at the start of 20X7, Norzani switched over to double-declining-balance (DDB) depreciation. 20X7 is expected to be the same as 20X6 except for the change in depreciation method. If Norzani had been using DDB depreciation all along, how much net income can Norzani, Inc., expect to earn during 20X7? Ignore income tax.

E7-46. (Learning Objective 1: Capitalizing versus expensing; measuring the effect of an error) All French Press (AFP) is a major French telecommunication conglomerate. Assume that early in year 1, AFP purchased equipment at a cost of €8.4 million. Management expects the equipment to remain in service for four years and the estimated residual value to be

L0 1

negligible. AFP uses the straight-line depreciation method. *Through an accounting error, AFP expensed the entire cost of the equipment at the time of purchase.* Because AFP is operated as a partnership, it pays no income tax.

Requirements

Prepare a schedule to show the overstatement or understatement in the following items at the end of each year over the four-year life of the equipment:

1. Total current assets
2. Equipment, net
3. Net income

Quiz

Test your understanding of accounting for PPE, natural resources, and intangibles by answering the following questions. Select the best choice from among the possible answers given.

Q7-47. A capital expenditure

- | | |
|-----------------------------|---|
| a. is expensed immediately. | c. records additional capital. |
| b. adds to an asset. | d. is a credit like capital (owners' equity). |

Q7-48. Which of the following items should be accounted for as a capital expenditure?

- a. Costs incurred to repair leaks in the building roof.
- b. Maintenance fees paid with funds provided by the company's capital.
- c. Taxes paid in conjunction with the purchase of office equipment.
- d. The monthly rental cost of an office building.

Q7-49. Suppose you buy land for \$3,200,000 and spend \$1,300,000 to develop the property. You then divide the land into lots as follows:

Category	Sale Price per Lot
10 hilltop lots.....	\$525,000
10 valley lots.....	350,000

How much did each hilltop lot cost you?

- | | |
|--------------|--------------|
| a. \$270,000 | c. \$128,000 |
| b. \$192,000 | d. \$180,000 |

Q7-50. Which statement about depreciation is false?

- a. Obsolescence as well as physical wear and tear should be considered when determining the period over which an asset should be depreciated.
- b. Depreciation is a process of allocating the cost of an asset to expense over its useful life.
- c. A major objective of depreciation accounting is to match the cost of using an asset with the revenues it helps to generate.
- d. Depreciation should not be recorded in years that the market value of the asset has increased.

Q7-51. Cleveland Corporation acquired a machine for \$42,000 and has recorded depreciation for two years using the straight-line method over a five-year life and \$7,000 residual value. At the start of the third year of use, Cleveland revised the estimated useful life to a total of 10 years. Estimated residual value declined to \$0.

What is the book value of the machine at the end of two full years of use?

- | | |
|-------------|-------------|
| a. \$21,000 | c. \$14,000 |
| b. \$22,000 | d. \$28,000 |

Q7-52. Cleveland Corporation acquired a machine for \$42,000 and has recorded depreciation for two years using the straight-line method over a five-year life and \$7,000 residual value. At the start of the third year of use, Cleveland revised the estimated useful life to a total of 10 years. Estimated residual value declined to \$0.

How much depreciation should Cleveland record in each of the asset's last eight years (that is, year 3 through year 10), following the revision?

- a. \$9,333
- b. \$2,800
- c. \$3,500
- d. Some other amount

Q7-53. King Company failed to record depreciation of equipment. How does this omission affect King's financial statements?

- a. Net income is understated, and assets are overstated.
- b. Net income is overstated, and assets are understated.
- c. Net income is understated, and assets are understated.
- d. Net income is overstated, and assets are overstated.

Q7-54. Jimmy's DVD, Inc., uses the double-declining-balance method for depreciation on its computers. Which item is not needed to compute depreciation for the first year?

- a. Original cost
- b. Estimated residual value
- c. Expected useful life in years
- d. All the above are needed.

Q7-55. Which of the following costs is reported on a company's Income Statement?

- a. Accumulated depreciation
- b. Land
- c. Accounts payable
- d. Depreciation expense

Q7-56. Which of the following items is reported on the Balance Sheet?

- a. Accumulated depreciation
- b. Gain on disposal of equipment
- c. Cost of goods sold
- d. Net sales revenue

Use the following information to answer questions 7-57 through 7-59.

Hal Company purchased a machine for \$9,500 on January 1, 20X6. The machine has been depreciated using the straight-line method over a 10-year life and \$500 residual value. Hal sold the machine on January 1, 20X8, for \$8,200.

Q7-57. What gain or loss should Hal record on the sale?

- a. Gain, \$500
- b. Loss, \$400
- c. Gain, \$1,000
- d. Gain, \$100

Q7-58. Journalize Hal's sale of the machine.

Q7-59. What is straight-line depreciation for the year ended December 31, 20X6, and what is the book value on December 31, 20X7?

Q7-60. A company purchased mineral assets costing \$960,000, with an estimated residual value of \$40,000, holding approximately 400,000 tons of ore. During the first year, 48,000 tons are extracted and sold. What is the amount of depletion for the first year?

- a. \$110,400
- b. \$96,000
- c. \$115,200
- d. Cannot be determined from the data given

Q7-61. Suppose Smooth Delivery pays \$68 million to buy Guaranteed Overnight. Guaranteed's assets are valued at \$74 million, and its liabilities total \$16 million. How much goodwill did Timely Delivery purchase in its acquisition of Guaranteed Overnight?

- a. \$22 million
- b. \$16 million
- c. \$10 million
- d. \$6 million

Problems MyLab Accounting

Select A and B problems can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select problems using MyLab Accounting.

Group A

P7-62A. (*Learning Objectives 1, 2: Identifying the elements of a PPE's cost; depreciation*)

Assume Online, Inc., opened an office in Durban, South Africa. Further assume that Online

incurred the following costs in acquiring land, making land improvements, and constructing and furnishing the new building:

a. Purchase price of land, including an old building that will be used for a garage (land market value is \$315,000; building market value is \$85,000).....	\$380,000
b. Landscaping (additional dirt and earth moving).....	8,700
c. Fence around the land.....	31,600
d. Attorney fee for title search on the land.....	900
e. Delinquent real estate taxes on the land to be paid by Online.....	5,700
f. Company signs at entrance to the property.....	1,300
g. Building permit for the sales building.....	500
h. Architect fee for the design of the sales building.....	19,700
i. Masonry, carpentry, and roofing of the sales building.....	525,000
j. Renovation of the garage building.....	41,300
k. Interest cost on construction loan for sales building.....	9,200
l. Landscaping (trees and shrubs).....	6,700
m. Parking lot and concrete walks on the property.....	53,100
n. Lights for the parking lot and walkways.....	7,600
o. Salary of construction supervisor (86% to sales building; 11% to land improvements; and 3% to garage building renovations).....	45,000
p. Office furniture for the sales building.....	79,900
q. Transportation and installation of furniture.....	1,800

Assume Online depreciates buildings over 40 years, land improvements over 20 years, and furniture over 10 years, all on a straight-line basis with zero residual value.

Requirements

1. Show how to account for each of Online's costs by listing the cost under the correct account. Determine the total cost of each asset.
2. All construction was complete and the assets were placed in service on May 2. Record depreciation for the year ended December 31. Round to the nearest dollar.
3. How will what you learned in this problem help you manage a business?

L0 1 2

P7-63A. (*Learning Objectives 1, 2: Recording PPE transactions; reporting on the Balance Sheet*) Rocco Lakes Resort reported the following on its Balance Sheet at December 31, 20X6:

Property, plant, and equipment, at cost:	
Land.....	\$ 146,000
Buildings.....	709,000
Less: Accumulated depreciation.....	(342,000)
Equipment.....	405,000
Less: Accumulated depreciation.....	(265,000)

In early July 20X7, the resort expanded operations and purchased additional equipment at a cost of \$108,000. The company depreciates buildings by the straight-line method over 20 years with residual value of \$84,000. Due to obsolescence, the equipment has a useful life of only 10 years and is being depreciated by the double-declining-balance method with zero residual value.

Requirements

1. Journalize Rocco Lakes Resort's PPE purchase and depreciation transactions for 20X7.
2. Report PPEs on the December 31, 20X7, Balance Sheet.

P7-64A. (Learning Objectives 1, 2, 3: Recording PPE transactions; exchanges; and changes in useful life) Carr, Inc., has the following PPE accounts: Land, Buildings, and Equipment, with a separate accumulated depreciation account for each of these except land. Carr completed the following transactions:

L0 1 2 3

Jan 2	Traded in equipment with accumulated depreciation of \$65,000 (cost of \$136,000) for similar new equipment with a cash cost of \$175,000. Received a trade-in allowance of \$75,000 on the old equipment and paid \$100,000 in cash.
Jun 30	Sold a building that had a cost of \$655,000 and had accumulated depreciation of \$130,000 through December 31 of the preceding year. Depreciation is computed on a straight-line basis. The building has a 40-year useful life and a residual value of \$275,000. Carr received \$115,000 cash and a \$405,250 note receivable.
Oct 29	Purchased land and a building for a single price of \$390,000. An independent appraisal valued the land at \$221,100 and the building at \$180,900.
Dec 31	Recorded depreciation as follows: Equipment has an expected useful life of 5 years and an estimated residual value of 6% of cost. Depreciation is computed on the double-declining-balance method. Depreciation on buildings is computed by the straight-line method. The new building carries a 40-year useful life and a residual value equal to 20% of its cost.

Requirement

1. Record the transactions in Carr, Inc.'s journal.

P7-65A. (Learning Objective 1: Explaining the concept of depreciation) The board of directors of Crystal Structures, Inc., is reviewing the 20X6 annual report. A new board member—a wealthy woman with little business experience—questions the company accountant about the depreciation amounts. The new board member wonders why depreciation expense has decreased from \$240,000 in 20X4 to \$224,000 in 20X5 to \$216,000 in 20X6. She states that she could understand the decreasing annual amounts if the company had been disposing of properties each year, but that has not occurred. Further, she notes that growth in the city is increasing the values of company properties. Why is the company recording depreciation when the property values are increasing?

L0 1

■ writing assignment

P7-66A. (Learning Objectives 1, 2, 3: Computing depreciation by three methods; identifying the cash-flow advantage of accelerated depreciation for tax purposes; subsequent costs) On January 9, 20X6, J. T. Orlando Co. paid \$240,000 for a computer system. In addition to the basic purchase price, the company paid a setup fee of \$2,000, \$8,000 sales tax, and \$30,000 for a special platform on which to place the computer. J. T. Orlando management estimates that the computer will remain in service for five years and have a residual value of \$20,000. The computer will process 30,000 documents the first year, with annual processing decreasing by 2,500 documents during each of the next four years (that is, 27,500 documents in year 20X7; 25,000 documents in year 20X8; and so on). In trying to decide which depreciation method to use, the company president has requested a depreciation schedule for each of the three depreciation methods (straight-line, units-of-production, and double-declining-balance).

L0 1 2 3

■ spreadsheet

Requirements

1. For each of the generally accepted depreciation methods, prepare a depreciation schedule showing asset cost, depreciation expense, accumulated depreciation, and asset book value.
2. J. T. Orlando reports to shareholders and creditors in the financial statements using the depreciation method that maximizes reported income in the early years of asset use. For income tax purposes, the company uses the depreciation method that minimizes income tax payments in those early years. Consider the first year J. T. Orlando Co. uses the computer. Identify the depreciation methods that meet Orlando's objectives, assuming the income tax authorities permit the use of any of the methods.

L0 2 3 6

P7-67A. (*Learning Objectives 2, 3, 6: Depreciation; recording intangibles; analyzing PPE transactions from a company's financial statements*) Gadgets, Inc., sells electronics and appliances. The excerpts that follow are adapted from Gadgets's financial statements for 20X6 and 20X5.

Balance Sheet (dollars in millions)	February 28	
	20X6	20X5
Assets		
Total current assets	\$7,980	\$6,900
Property, plant, and equipment	4,835	4,198
Less: Accumulated depreciation	2,128	1,724
Goodwill	560	520

Statement of Cash Flows (dollars in millions)	Year Ended February 28	
	20X6	20X5
Operating activities:		
Net income	\$1,146	\$ 981
Non-cash items affecting net income:		
Depreciation	460	457
Investing activities:		
Additions to property, plant, and equipment	(707)	(615)

Requirements

1. How much was Gadgets's cost of PPE at February 28, 20X6? How much was the book value of PPE? Show computations.
2. The financial statements give three pieces of evidence that Gadgets purchased PPE and goodwill during fiscal year 20X6. What are they?
3. Prepare T-accounts for Property, Plant, and Equipment; Accumulated Depreciation; and Goodwill. Then show all the activity in these accounts during 20X6. Label each increase or decrease and give its dollar amount. During 20X6, Gadgets sold PPE that had cost the company \$70 million (accumulated depreciation on these assets was \$56 million). Assume there were no losses on goodwill during 20X6.

L0 4

P7-68A. (*Learning Objective 4: Accounting for natural resources, and the related expense*)

NorthAtlantic Energy Company's Balance Sheet includes the asset Iron Ore. NorthAtlantic Energy paid \$2.8 million cash for a lease giving the firm the right to work a mine that contained an estimated 198,000 tons of ore. The company paid \$68,000 to remove unwanted buildings from the land and \$72,000 to prepare the surface for mining. NorthAtlantic Energy also signed a \$37,300 note payable to a landscaping company to return the land surface to its original condition after the lease ends. During the first year, NorthAtlantic Energy removed 34,500 tons of ore, which it sold on account for \$36 per ton. Operating expenses for the first year totaled \$250,000, all paid in cash. In addition, the company accrued income tax at the tax rate of 32%.

Requirements

1. Record all of NorthAtlantic Energy's transactions for the year.
2. Prepare the company's Income Statement for its iron ore operations for the first year. Evaluate the profitability of the company's operations.

P7-69A. (Learning Objectives 4, 6: Reporting PPE transactions on the statement of cash flows) At the end of 20X5, Solving Engineering Associates (SEA) had total assets of \$17.1 billion and total liabilities of \$9.7 billion. Included among the assets were property, plant, and equipment with a cost of \$4.4 billion and accumulated depreciation of \$3.2 billion.

LO 4 6

SEA completed the following selected transactions during 20X6. The company earned total revenues of \$26.6 billion and incurred total expenses of \$21.2 billion, which included depreciation of \$1.9 billion. During the year, SEA paid \$1.8 billion for new property, plant, and equipment and sold old PPE for \$0.4 billion. The cost of the assets sold was \$1.2 billion, and their accumulated depreciation was \$0.6 billion.

Requirements

1. Explain how to determine whether SEA had a gain or loss on the sale of old PPE during the year. What was the amount of the gain or loss, if any?
2. Show how SEA would report property, plant, and equipment on the Balance Sheet at December 31, 20X6, after all the year's activity. What was the book value of property, plant, and equipment?
3. Show how SEA would report its operating activities and investing activities on its statement of cash flows for 20X6. Ignore gains and losses.

Group B

P7-70B. (Learning Objectives 1, 2: Identifying the elements of a PPE's cost) Assume Lance Pharmacy, Inc., opened an office in Valencia, Italy. Further assume that Lance Pharmacy incurred the following costs in acquiring land, making land improvements, and constructing and furnishing the new sales building:

LO 1 2

a. Purchase price of land, including an old building that will be used for a garage (land market value is €310,000; building market value is €90,000).....	€320,000
b. Landscaping (additional dirt and earth moving).....	8,200
c. Fence around the land.....	32,000
d. Attorney fee for title search on the land.....	300
e. Delinquent real estate taxes on the land to be paid by Lance Pharmacy.....	5,800
f. Company signs at entrance to the property.....	1,500
g. Building permit for the sales building.....	800
h. Architect fee for the design of the sales building.....	19,800
i. Masonry, carpentry, and roofing of the sales building.....	505,000
j. Renovation of the garage building.....	41,800
k. Interest cost on construction loan for sales building.....	9,400
l. Landscaping (trees and shrubs).....	6,500
m. Parking lot and concrete walks on the property.....	52,800
n. Lights for the parking lot and walkways.....	7,100
o. Salary of construction supervisor (86% to sales building; 10% to land improvements; and 4% to garage building renovations).....	40,000
p. Office furniture for the sales building.....	78,200
q. Transportation and installation of furniture.....	1,200

Assume Lance Pharmacy depreciates buildings over 30 years, land improvements over 15 years, and furniture over eight years, all on a straight-line basis with zero residual value.

Requirements

1. Show how to account for each of Lance Pharmacy's costs by listing the cost under the correct account. Determine the total cost of each asset.
2. All construction was complete and the assets were placed in service on May 2. Record depreciation for the year ended December 31. Round to the nearest dollar.
3. How will what you learned in this problem help you manage a business?

L0 2 3

P7-71B. (*Learning Objectives 2, 3: Recording PPE transactions; reporting on the Balance Sheet*) Pokko Lakes Resort reported the following on its Balance Sheet at December 31, 20X6:

Property, plant, and equipment, at cost:	
Land.....	€ 149,000
Buildings	704,000
Less: Accumulated depreciation	(342,000)
Equipment.....	401,000
Less: Accumulated depreciation	(268,000)

In early July 20X7, the resort expanded operations and purchased additional equipment at a cost of €104,000. The company depreciates buildings by the straight-line method over 20 years with residual value of €88,000. Due to obsolescence, the equipment has a useful life of only 10 years and is being depreciated by the double-declining-balance method with zero residual value.

Requirements

1. Journalize Pokko Lakes Resort's PPE purchase and depreciation transactions for 20X7.
2. Report PPE on the December 31, 20X7, Balance Sheet.

L0 1 2 3

P7-72B. (*Learning Objectives 1, 2, 3: Recording PPE transactions, exchanges, and changes in useful life*) Tarrier, Inc., has the following PPE accounts: Land, Buildings, and Equipment, with a separate accumulated depreciation account for each of these except land. Tarrier completed the following transactions:

Jan 2	Traded in equipment with accumulated depreciation of €64,000 (cost of €138,000) for similar new equipment with a cash cost of €179,000. Received a trade-in allowance of €73,000 on the old equipment and paid €106,000 in cash.
Jun 30	Sold a building that had a cost of €645,000 and had accumulated depreciation of €155,000 through December 31 of the preceding year. Depreciation is computed on a straight-line basis. The building has a 40-year useful life and a residual value of €285,000. Tarrier received €135,000 cash and a €350,500 note receivable.
Oct 29	Purchased land and a building for a single price of €340,000. An independent appraisal valued the land at €108,900 and the building at €254,100.
Dec 31	Recorded depreciation as follows: Equipment has an expected useful life of 4 years and an estimated residual value of 4% of cost. Depreciation is computed on the double-declining-balance method. Depreciation on buildings is computed by the straight-line method. The new building carries a 40-year useful life and a residual value equal to 10% of its cost.

Requirement

1. Record the transactions in Tarrier, Inc.'s journal.

L0 2

■ writing assignment

P7-73B. (*Learning Objective 2: Explaining the concept of depreciation*) The board of directors of Looper Structures, Inc., is reviewing the 20X6 annual report. A new board member—a wealthy man with little business experience—questions the company accountant about the depreciation amounts. The new board member wonders why depreciation expense has decreased from €200,000 in 20X4 to €184,000 in 20X5 to €176,000 in 20X6. He states that he could understand the decreasing annual amounts if the company had been disposing of properties each year, but that has not occurred. Further, he notes that growth in the city is increasing the values of company properties. Why is the company recording depreciation when the property values are increasing?

P7-74B. (*Learning Objectives 1, 2, 3: Computing depreciation by three methods; identifying the cash-flow advantage of accelerated depreciation for tax purposes*) On January 6, 20X6, K. P. Scott Co. paid €265,000 for a computer system. In addition to the basic purchase price, the company paid a setup fee of €800, €6,400 sales tax, and €27,800 for a special platform on which to place the computer. K. P. Scott management estimates that the computer will remain in service for five years and have a residual value of €30,000. The computer will process 45,000 documents the first year, with annual processing decreasing by 2,500 documents during each of the next four years (that is, 42,500 documents in 20X7; 40,000 documents in 20X8; and so on). In trying to decide which depreciation method to use, the company president has requested a depreciation schedule for each of the three depreciation methods (straight-line, units-of-production, and double-declining-balance).

LO 1 2 3

■ spreadsheet

Requirements

1. For each of the generally accepted depreciation methods, prepare a depreciation schedule showing asset cost, depreciation expense, accumulated depreciation, and asset book value.
2. K. P. Scott reports to shareholders and creditors in the financial statements using the depreciation method that maximizes reported income in the early years of asset use. For income tax purposes, the company uses the depreciation method that minimizes income tax payments in those early years. Consider the first year K. P. Scott Co. uses the computer. Identify the depreciation methods that meet Scott's objectives, assuming the income tax authorities permit the use of any of the methods.

P7-75B. (*Learning Objectives 2, 3, 6: Depreciation; recording intangibles; analyzing PPE transactions from a company's financial statements*) Perma, Inc., sells electronics and appliances. The excerpts that follow are adapted from Perma's financial statements for 20X6 and 20X5.

LO 2 3 6

Balance Sheet (Euros in millions)	February 28	
	20X6	20X5
Assets		
Total current assets	€7,988	€6,905
Property, plant, and equipment	4,830	4,192
Less: Accumulated depreciation	2,129	1,728
Goodwill	552	514

Statement of Cash Flows (Euros in millions)	Year Ended February 28	
	20X6	20X5
Operating activities:		
Net income	€1,147	€ 989
Non-cash items affecting net income:		
Depreciation	458	460
Investing activities:		
Additions to property, plant, and equipment	(716)	(617)

Requirements

1. How much was Perma's cost of PPE at February 28, 20X6? How much was the book value of PPE? Show computations.
2. The financial statements give three pieces of evidence that Perma purchased PPE and goodwill during fiscal year 20X6. What are they?

3. Prepare T-accounts for Property, Plant, and Equipment; Accumulated Depreciation; and Goodwill. Then show all the activity in these accounts during 20X6. Label each increase or decrease and give its dollar amount. During 20X6, Perma sold PPE that had cost the company €78 million (accumulated depreciation on these assets was €57 million). Assume there were no losses on goodwill during 20X6.

L0 4**P7-76B. (Learning Objective 4: Accounting for natural resources and the related expense)**

South Pacific Energy Company's Balance Sheet includes the asset Iron Ore. South Pacific Energy paid €2.2 million cash for a lease giving the firm the right to work a mine that contained an estimated 192,000 tons of ore. The company paid €64,600 to remove unwanted buildings from the land and €71,000 to prepare the surface for mining. South Pacific Energy also signed a €26,000 note payable to a landscaping company to return the land surface to its original condition after the lease ends. During the first year, South Pacific Energy removed 31,500 tons of ore, which it sold on account for €32 per ton. Operating expenses for the first year totaled €242,000, all paid in cash. In addition, the company accrued income tax at the tax rate of 25%.

Requirements

1. Record all of South Pacific Energy's transactions for the year.
2. Prepare the company's Income Statement for its iron ore operations for the first year. Evaluate the profitability of the company's operations.

L0 6**P7-77B. (Learning Objective 6: Reporting PPE transactions on the statement of cash flows)**

At the end of 20X5, Great Financial Associates (GFA) had total assets of €17.4 billion and total liabilities of €9.9 billion. Included among the assets were property, plant, and equipment with a cost of €4.6 billion and accumulated depreciation of €3.3 billion.

GFA completed the following selected transactions during 20X6. The company earned total revenues of €26.1 billion and incurred total expenses of €21.0 billion, which included depreciation of €1.9 billion. During the year, GFA paid €1.6 billion for new property, plant, and equipment and sold old PPE for €0.3 billion. The cost of the assets sold was €1.1 billion, and their accumulated depreciation was €0.5 billion.

Requirements

1. Explain how to determine whether GFA had a gain or loss on the sale of old PPE during the year. What was the amount of the gain or loss, if any?
2. Show how GFA would report property, plant, and equipment on the Balance Sheet at December 31, 20X6, after all the year's activity. What was the book value of property, plant, and equipment?
3. Show how GFA would report its operating activities and investing activities on its statement of cash flows for 20X6. Ignore gains and losses.

APPLY YOUR KNOWLEDGE

Decision Cases

L0 2 3**■ writing assignment****Case 1. (Learning Objectives 2, 3: Measuring profitability based on different inventory and depreciation methods)**

Suppose you are considering investing in two businesses, La Petite France Bakery and Burgers Ahoy. The two companies are virtually identical, and both began operations at the beginning of the current year. During the year, each company purchased inventory as follows:

Jan	4	10,000 units at \$4 =	40,000
Apr	6	5,000 units at \$5 =	25,000
Aug	9	7,000 units at \$6 =	42,000
Nov	27	10,000 units at \$7 =	70,000
Totals		<u>32,000</u>	<u>\$177,000</u>

During the first year, both companies sold 28,000 units of inventory.

In early January, both companies purchased equipment costing \$180,000 that had a 10-year estimated useful life and a \$30,000 residual value. La Petite France uses the inventory and depreciation methods that maximize reported income. By contrast, Burgers uses the inventory and depreciation methods that minimize income tax payments. Assume that both companies' trial balances at December 31 included the following:

Sales revenue	\$350,000
Operating expenses	50,000

The income tax rate is 35%.

Requirements

1. Prepare both companies' Income Statements.
2. Write an investment newsletter to address the following questions: Which company appears to be more profitable? Which company has more cash to invest in promising projects? If prices continue rising over the long term, which company would you prefer to invest in? Why? (Challenge)

Case 2. (*Learning Objectives 1, 5: Accounting for PPE and intangible assets*) The following questions are unrelated except that they all apply to PPEs and intangible assets:

1. The manager of Copper World regularly debits the cost of repairs and maintenance of PPE to Plant and Equipment. Why would she do that, since she knows she is violating GAAP?
2. The manager of Hillion Software regularly buys PPE and debits the cost to Repairs and Maintenance Expense. Why would he do that, since he knows this action violates GAAP?
3. It has been suggested that because many intangible assets have no value except to the company that owns them, they should be valued at \$1.00 or zero on the Balance Sheet. Many accountants disagree with this view. Which view do you support? Why?

LO 1 5

■ writing assignment

Ethical Issue

United Jersey Bank of Princeton purchased land and a building for the lump sum of \$6.0 million. To get the maximum tax deduction, the bank's managers allocated 80% of the purchase price to the building and only 20% to the land. A more realistic allocation would have been 60% to the building and 40% to the land.



Requirements

1. What is the ethical issue in this situation?
2. Who are the stakeholders? What are the possible consequences to each?
3. Analyze the alternatives from the following standpoints: (a) economic, (b) legal, and (c) ethical.
4. What would you do? How would you justify your decision?

Focus on Financials: | Nestlé

This case spans all 12 chapters and is based on the consolidated financial statements of Nestlé. As you work with Nestlé throughout this course, you will develop the confidence and ability to use the financial statements of other companies as well. Refer to Nestlé's consolidated financial statements in Appendix A. If you wish, you can obtain the full annual report from www.nestle.com/investors. You may find the information overwhelming for now, but try to spot the key principles that we have discussed in this chapter. It will get progressively easier as you gain familiarity with the elements of the financial statements.



Requirements

1. Refer to Note 8—Property, plant and equipment. What kinds of fixed assets does Nestlé have?
2. What depreciation method does Nestlé use for reporting to shareholders and creditors in the financial statements? Assume that Nestlé expects fairly constant net income before

depreciation for the next few years. What type of depreciation method would the company probably use for income tax purposes and why?

3. Depreciation expense is not disclosed separately on Nestlé's Consolidated Income Statement. Instead, it is lumped together with other expenses. Nevertheless, we can find the separate figure from the Consolidated Statements of Cash Flows. What was Nestlé's depreciation and amortization expense during 2016? (You can refer to Note 8.) From that note, what was Nestlé's accumulated depreciation at the end of 2016? Ignore the impairment aspect of this figure and treat it as immaterial. Explain why accumulated depreciation exceeds depreciation expense for the current year.
4. What are Nestlé's intangible assets? Assuming straight line amortization of finite live intangible assets, what are your estimates for Nestlé's intangible assets?
5. Refer to Note 8. How much impairment of PPE did Nestlé record for 2016? What do you understand by "impairment"? Examine Note 8 carefully and describe a typical event that would lead to an impairment of PPE.

Group Project

Visit a local business.

Requirements

1. List all its PPE.
2. If possible, interview the manager. Gain as much information as you can about the business's PPE. For example, try to determine the assets' costs, the depreciation method the company is using, and the estimated useful life of each asset category. If an interview is impossible, then develop your own estimates of the assets' costs, useful lives, and book values, assuming an appropriate depreciation method.
3. Determine whether the business has any intangible assets. If so, list them and gain as much information as possible about their nature, cost, and estimated useful lives.
4. Write a detailed report of your findings and be prepared to present your results to the class.

Quick Check Answers

1. $b \{ [\$1,420/(\$1,420 + \$650 + \$430)] \times \$1,800 = \$1,022.4 \}$
2. c
3. $a \{ (\$65,000 - \$5,000)/10 \times 6/12 = \$3,000 \}$
4. $d \{ [(\$65,000 - \$5,000)/10 \times 2 = \$12,000; \$65,000 - \$12,000 = \$53,000] \}$
5. $b \{ \$65,000 \times .2 = \$13,000; (\$65,000 - \$13,000) \times .2 = \$10,400 \}$
6. $a \{ [(\$65,000 - \$5,000)/10 \times 4 = \$24,000; \$65,000 - \$24,000 = \$41,000; \$44,500 - \$41,000 = \text{gain of } \$3,500] \}$
7. $c \{ \$30,000 \times 2/5 = \$12,000; (\$30,000 - \$12,000) \times 2/5 = \$7,200; \$12,000 + \$7,200 = \$19,200 \}$
8. e
9. a
10. d
11. $c \{ \$270,000 \times (20,000/100,000) = \$54,000 \}$
12. e

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